Assignment 3

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Verification Diploma

Q1

1. Testbench code

```
import testing_pkg::*;

module tb();

  Transaction tr = new();
  byte operand1, operand2, out, expected;
  opcode_e opcode;
  logic clk, rst;

  int correct_count = 0;
  int error_count   = 0;

  // Clock Generation
  initial begin
    clk = 0;
    forever begin
      #5 clk = ~clk;
      tr.clk = clk;
    end
  end

  // DUT Instantiation
  alu_seq DUT (
    .operand1 (operand1),
    .operand2 (operand2),
    .clk      (clk),
    .rst      (rst),
    .opcode   (opcode),
    .out      (out)
  );

  // ---------------------------
  // Tasks
  // ---------------------------
```

```
36    // Reset task
37    task do_reset();
38      rst = 1;
39      @(negedge clk);
40      rst = 0;
41    endtask
42
43    // Calculate expected output
44    task calc_expected();
45      case (opcode)
46        ADD:  expected = operand1 + operand2;
47        SUB:  expected = operand1 - operand2;
48        MULT: expected = operand1 * operand2;
49        DIV:  expected = (operand2 != 0) ? (operand1 / operand2) : 0;
50        default: expected = 0;
51      endcase
52    endtask
53
54    // Compare DUT vs Expected
55    task check_output();
56      if (out === expected) begin
57        correct_count++;
58        $display("PASS: opcode=%0s, op1=%0d, op2=%0d, out=%0d",
59                 opcode.name(), operand1, operand2, out);
60      end
61      else begin
62        error_count++;
63        $display("FAIL: opcode=%0s, op1=%0d, op2=%0d, expected=%0d, got=%0d",
64                 opcode.name(), operand1, operand2, expected, out);
65      end
66    endtask
```

```
68    task force_operand1_operand2_cross();
69    static byte corners[3] = '{-128, 0, 127};
70    int i, j;
71
72    for (i = 0; i < 3; i++) begin
73      for (j = 0; j < 3; j++) begin
74        operand1 = corners[i];
75        operand2 = corners[j];
76        opcode   = ADD;
77        opcode   = SUB;
78        opcode   = MULT;
79        opcode   = DIV;
80
81        @(negedge clk);
82        calc_expected();
83        check_output();
84      end
85    end
86  endtask
```

```
89      // ---------------------------
90      // Stimulus
91      // ---------------------------
92      initial begin
93        do_reset();
94
95        force_operand1_operand2_cross();
96
97        repeat (5000) begin
98          assert(tr.randomize());  // generate transaction
99          operand1 = tr.operand1;
100         operand2 = tr.operand2;
101         opcode   = tr.opcode;
102
103         @(negedge clk); // wait for DUT
104         calc_expected();
105         check_output();
106
107         do_reset();
108       end
109
110       $display("=======================================");
111       $display("Simulation finished!");
112       $display("Correct Count = %0d", correct_count);
113       $display("Error Count   = %0d", error_count);
114       $display("=======================================");
115
116       $stop();
117     end
118   endmodule
```

2. Package code

```
package testing_pkg;

  typedef enum {ADD, SUB, MULT, DIV} opcode_e;

  class Transaction;
    rand opcode_e opcode;
    rand byte operand1;
    rand byte operand2;
    bit clk;

    covergroup CovCode @(posedge clk);

      operand1_cp: coverpoint operand1 {
        bins max_neg = {-128};
        bins zero    = {0};
        bins max_pos = {127};
        bins misc    = default;
      }


    operand2_cp: coverpoint operand2 {
        bins max_neg = {-128};
        bins zero    = {0};
        bins max_pos = {127};
        bins misc    = default;
      }


      opcode_cp: coverpoint opcode {
        bins add_sub     = {ADD, SUB};
    bins add_mult     = {ADD, MULT};
        bins mult_sub     = {MULT, SUB};
        bins add_then_sub = (ADD => SUB);
    bins add_then_mult = (SUB => MULT);
    bins mult_then_sub = (MULT => ADD);
        illegal_bins no_div = {DIV};
```

```
36          illegal_bins no_div = {DIV};
37        }
38
39      opcode_operand1: cross opcode_cp, operand1_cp {
40          option.weight = 5;
41          option.cross_auto_bin_max = 0;
42
43          bins max_pos_add_or_sub =
44           binsof(operand1_cp.max_pos) && binsof(opcode_cp.add_sub);
45
46      bins max_pos_add_or_mult =
47           binsof(operand1_cp.max_pos) && binsof(opcode_cp.add_mult);
48
49      bins max_pos_mult_or_sub =
50           binsof(operand1_cp.max_pos) && binsof(opcode_cp.mult_sub);
51
52          bins max_neg_add_or_sub =
53           binsof(operand1_cp.max_neg) && binsof(opcode_cp.add_sub);
54
55      bins max_neg_add_or_mult =
56           binsof(operand1_cp.max_neg) && binsof(opcode_cp.add_mult);
57
58      bins max_neg_mult_or_sub =
59           binsof(operand1_cp.max_neg) && binsof(opcode_cp.mult_sub);
60        }
61
62
63      opcode_operand2: cross opcode_cp, operand2_cp {
64          option.weight = 5;
65          option.cross_auto_bin_max = 0;
66
67          bins max_pos_add_or_sub =
68           binsof(operand2_cp.max_pos) && binsof(opcode_cp.add_sub);
```

```
63      opcode_operand2: cross opcode_cp, operand2_cp {
64          option.weight = 5;
65          option.cross_auto_bin_max = 0;
66
67          bins max_pos_add_or_sub =
68          binsof(operand2_cp.max_pos) && binsof(opcode_cp.add_sub);
69
70      bins max_pos_add_or_mult =
71          binsof(operand2_cp.max_pos) && binsof(opcode_cp.add_mult);
72
73      bins max_pos_mult_or_sub =
74          binsof(operand2_cp.max_pos) && binsof(opcode_cp.mult_sub);
75
76          bins max_neg_add_or_sub =
77          binsof(operand2_cp.max_neg) && binsof(opcode_cp.add_sub);
78
79      bins max_neg_add_or_mult =
80          binsof(operand2_cp.max_neg) && binsof(opcode_cp.add_mult);
81
82      bins max_neg_mult_or_sub =
83          binsof(operand2_cp.max_neg) && binsof(opcode_cp.mult_sub);
84        }
85
86
87      endgroup
88
89      function new();
90        CovCode = new();
91      endfunction
92
93    endclass : Transaction
94
95  endpackage : testing_pkg
96
```

3. Design code

```
1   import testing_pkg::*;
2
3   module alu_seq(operand1, operand2, clk, rst, opcode, out);
4   input byte operand1, operand2;
5   input clk, rst;
6   input opcode_e opcode;
7   output byte out;
8
9   always @(posedge clk) begin
10    if (rst)
11      out <= 0;
12    else
13      case (opcode)
14        ADD: out <= operand1 + operand2;
15        SUB: out <= operand1 - operand2;
16        MULT:out <= operand1 * operand2;
17        DIV: out <= operand1 / operand2;
18        default: out <= 0;
19      endcase
20  end
21  endmodule
```

4. Report any bugs detected in the design and fix them: no any

5. Snippet to your verification plan document

Label	Design requirement description	Stimulus generation	Functional coverage	Functionality check
ALU_RESET	When reset is asserted, the output out should be zero.	Directed at start	Coverpoint on rst transition + out==0 bin	A checker task compares DUT output against expected=0 after reset.
ALU_ADD	For opcode=ADD, DUT output = operand1 + operand2.	Randomization + Directed corners	Coverpoint on opcode=ADD, cross with operand1_cp &	Checker task calculates expected ADD result and compares with DUT output.
ALU_SUB	For opcode=SUB, DUT output = operand1 - operand2.	Randomization + Directed corners	Coverpoint on opcode=SUB, cross with operand1_cp &	Checker task calculates expected SUB result and compares with DUT output.
ALU_MULT	For opcode=MULT, DUT output = operand1 * operand2.	Randomization	Coverpoint on opcode=MULT, operand1/operand2 directed to 0, 1, max_pos, max_neg	Checker task calculates expected MULT result and compares with DUT output.
ALU_DIV	For opcode=DIV, DUT output = operand1 / operand2 (excluding divide-by-zero).	Randomization + Directed (avoid div/0)	Coverpoint on opcode=DIV, illegal_bins for divide-by-zero	Checker task calculates expected DIV result and compares with DUT output.
ALU_OP1_OP2	Operand1 & Operand2 extreme values (−128, 0, 127) combinations must be exercised.	Directed 3×3 corner test	Cross operand1_cp × operand2_cp, bins for (max_pos, max_neg, zero) combinations	Checker ensures DUT output is consistent with operation for all corner cases.
ALU_OPCODE_SEQ	ALU should cover sequence ADD → SUB (temporal requirement).	Randomization	Coverpoint on opcode_cp.add_then_sub bin	Coverage ensures ADD is followed by SUB at least once; checker compares
ALU_NO_DIV	DIV opcode is marked illegal (by spec).	Constrained random	illegal_bins no_div ensures this bin never gets hit	Simulation must never see DIV when illegal; scoreboard check triggers if violated.

6. Do file

```
vlib work
vlog alu_seq.sv  testing_pkg.sv tb.sv  +cover -covercells
vsim -voptargs=+acc work.tb -cover
add wave *
coverage save tb.ucdb -onexit
run -all
```

7. Code Coverage report snippets (Questions 1, 2, and 3 only)

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```
Coverage Report by instance with details

================================================================================
=== Instance: /tb/DUT
=== Design Unit: work.alu_seq
================================================================================
Branch Coverage:
    Enabled Coverage              Bins      Hits    Misses  Coverage
    ----------------              ----      ----    ------  --------
    Branches                         6         6         0   100.00%

================================Branch Details================================

Branch Coverage for instance /tb/DUT

    Line         Item                      Count     Source
    ----         ----                      -----     ------
  File alu_seq.sv
-----------------------------------IF Branch------------------------------------
    10                                     10010     Count coming in to IF
    10              1                       5001        if (rst)

    12              1                       5009        else

Branch totals: 2 hits of 2 branches = 100.00%

---------------------------------CASE Branch------------------------------------
    13                                      5009     Count coming in to CASE
    14              1                       1240                         ADD: out <= operand1 + operand2;

    15              1                       1180                         SUB: out <= operand1 - operand2;

    16              1                       1232                         MULT:out <= operand1 * operand2;

    17              1                       1357                         DIV: out <= operand1 / operand2;

Branch totals: 4 hits of 4 branches = 100.00%
```

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```
Statement Coverage:
    Enabled Coverage              Bins      Hits    Misses  Coverage
    ----------------              ----      ----    ------  --------
    Statements                       6         6         0   100.00%

================================Statement Details================================

Statement Coverage for instance /tb/DUT --

    Line         Item                      Count     Source
    ----         ----                      -----     ------
  File alu_seq.sv
    3                                      	module alu_seq(operand1, operand2, clk, rst, opcode, out);
    4                                      	input byte operand1, operand2;
    5                                      	input clk, rst;
    6                                      	input opcode_e opcode;
    7                                      	output byte out;
    8
    9               1                  10010     always @(posedge clk) begin
    10                                     	   if (rst)
    11              1                   5001   	      out <= 0;
    12                                     	   else
    13                                     	      case (opcode)
    14              1                   1240   	         ADD: out <= operand1 + operand2;
    15              1                   1180   	         SUB: out <= operand1 - operand2;
    16              1                   1232   	         MULT:out <= operand1 * operand2;
    17              1                   1357   	         DIV: out <= operand1 / operand2;
```

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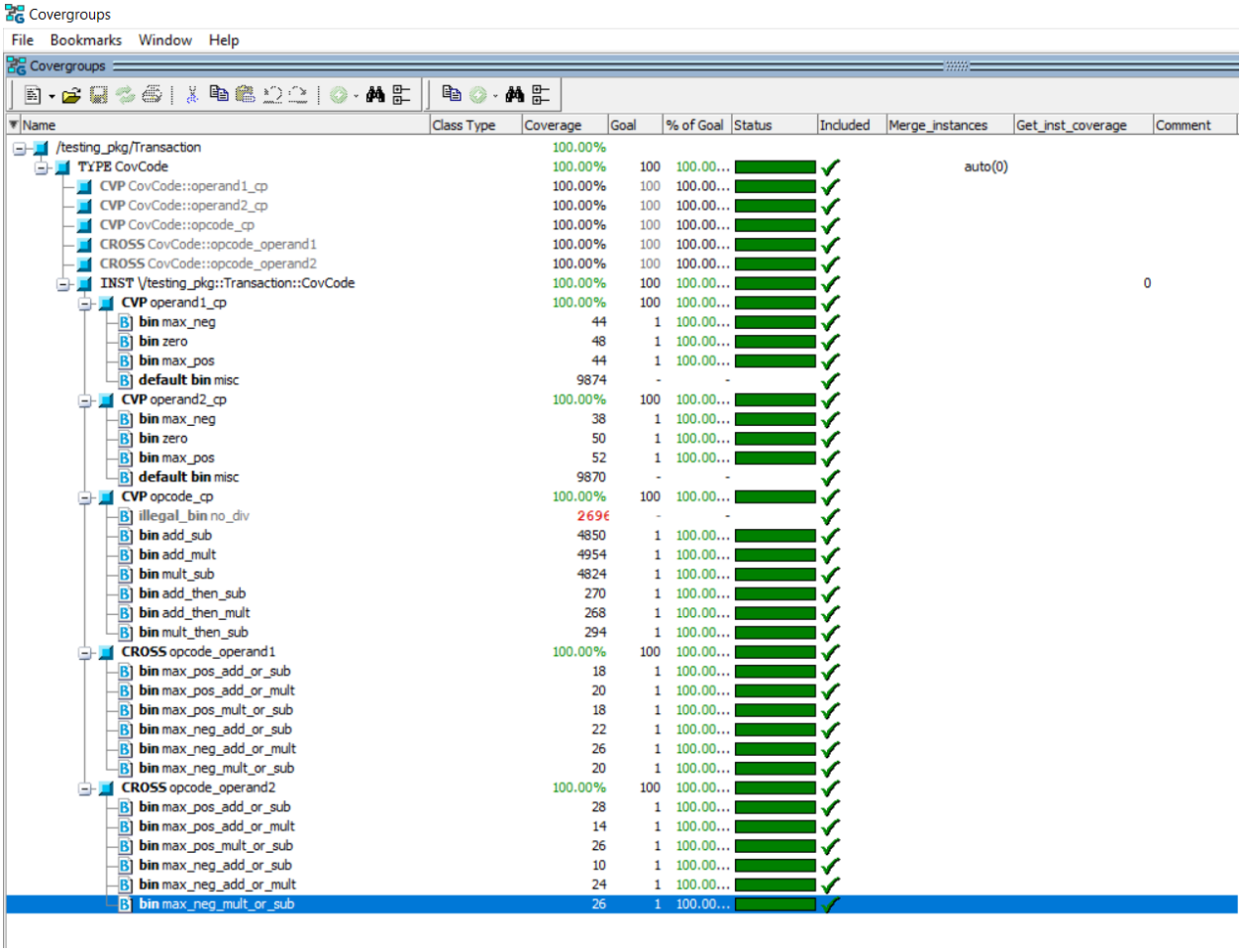
```
Toggle Coverage:
    Enabled Coverage              Bins      Hits    Misses  Coverage
    ----------------              ----      ----    ------  --------
    Toggles                         52        52         0   100.00%

================================Toggle Details================================

Toggle Coverage for instance /tb/DUT --

                                        Node      1H->0L      0L->1H                          "Coverage"
                                        ----------------------------------------------------------------
                                         clk           1           1                             100.00
                                operand1[7-0]          1           1                             100.00
                                operand2[7-0]          1           1                             100.00
                                     out[7-0]          1           1                             100.00
                                         rst           1           1                             100.00

Total Node Count     =         26
Toggled Node Count   =         26
Untoggled Node Count =          0

Toggle Coverage      =     100.00% (52 of 52 bins)

--------------------------------------------------------------------------------
```

8. Functional Coverage report snippets (Questions 1, 2, and 3 only)



Name	Class Type	Coverage	Goal	% of Goal	Status	Included	Merge_instances	Get_inst_coverage	Comment
/testing_pkg/Transaction		100.00%							
TYPE CovCode		100.00%	100	100.00...			auto(0)		
CVP CovCode::operand1_cp		100.00%	100	100.00...					
CVP CovCode::operand2_cp		100.00%	100	100.00...					
CVP CovCode::opcode_cp		100.00%	100	100.00...					
CROSS CovCode::opcode_operand1		100.00%	100	100.00...					
CROSS CovCode::opcode_operand2		100.00%	100	100.00...					
INST \/testing_pkg::Transaction::CovCode		100.00%	100	100.00...					0
CVP operand1_cp		100.00%	100	100.00...					
bin max_neg		44	1	100.00...					
bin zero		48	1	100.00...					
bin max_pos		44	1	100.00...					
default bin misc		9874	-	-					
CVP operand2_cp		100.00%	100	100.00...					
bin max_neg		38	1	100.00...					
bin zero		50	1	100.00...					
bin max_pos		52	1	100.00...					
default bin misc		9870	-	-					
CVP opcode_cp		100.00%	100	100.00...					
illegal_bin no_div		269€	-	-					
bin add_sub		4850	1	100.00...					
bin add_mult		4954	1	100.00...					
bin mult_sub		4824	1	100.00...					
bin add_then_sub		270	1	100.00...					
bin add_then_mult		268	1	100.00...					
bin mult_then_sub		294	1	100.00...					
CROSS opcode_operand1		100.00%	100	100.00...					
bin max_pos_add_or_sub		18	1	100.00...					
bin max_pos_add_or_mult		20	1	100.00...					
bin max_pos_mult_or_sub		18	1	100.00...					
bin max_neg_add_or_sub		22	1	100.00...					
bin max_neg_add_or_mult		26	1	100.00...					
bin max_neg_mult_or_sub		20	1	100.00...					
CROSS opcode_operand2		100.00%	100	100.00...					
bin max_pos_add_or_sub		28	1	100.00...					
bin max_pos_add_or_mult		14	1	100.00...					
bin max_pos_mult_or_sub		26	1	100.00...					
bin max_neg_add_or_sub		10	1	100.00...					
bin max_neg_add_or_mult		24	1	100.00...					
bin max_neg_mult_or_sub		26	1	100.00...					

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```
Covergroup Coverage:
    Covergroups                     1        na        na   100.00%
        Coverpoints/Crosses         5        na        na        na
            Covergroup Bins        24        24         0   100.00%
------------------------------------------------------------------------------------------------
Covergroup                                             Metric       Goal       Bins    Status

------------------------------------------------------------------------------------------------
 TYPE /testing_pkg/Transaction/CovCode                100.00%        100          -    Covered
    covered/total bins:                                    24         24          -
    missing/total bins:                                     0         24          -
    % Hit:                                             100.00%        100          -
    Coverpoint operand1_cp                             100.00%        100          -    Covered
        covered/total bins:                                 3          3          -
        missing/total bins:                                 0          3          -
        % Hit:                                         100.00%        100          -
    Coverpoint operand2_cp                             100.00%        100          -    Covered
        covered/total bins:                                 3          3          -
        missing/total bins:                                 0          3          -
        % Hit:                                         100.00%        100          -
    Coverpoint opcode_cp                               100.00%        100          -    Covered
        covered/total bins:                                 6          6          -
        missing/total bins:                                 0          6          -
        % Hit:                                         100.00%        100          -
    Cross opcode_operand1                              100.00%        100          -    Covered
        covered/total bins:                                 6          6          -
        missing/total bins:                                 0          6          -
        % Hit:                                         100.00%        100          -
    Cross opcode_operand2                              100.00%        100          -    Covered
        covered/total bins:                                 6          6          -
        missing/total bins:                                 0          6          -
        % Hit:                                         100.00%        100          -
 Covergroup instance \/testing_pkg::Transaction::CovCode
                                                       100.00%        100          -    Covered
    covered/total bins:                                    24         24          -
    missing/total bins:                                     0         24          -
    % Hit:                                             100.00%        100          -
    Coverpoint operand1_cp                             100.00%        100          -    Covered
        covered/total bins:                                 3          3          -
        missing/total bins:                                 0          3          -
        % Hit:                                         100.00%        100          -
        bin max_neg                                        44          1          -    Covered
        bin zero                                           48          1          -    Covered
```

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bin zero	48	1	-	Covered
bin max_pos	44	1	-	Covered
default bin misc	9874		-	Occurred
Coverpoint operand2_cp	100.00%	100	-	Covered
covered/total bins:	3	3	-	
missing/total bins:	0	3	-	
% Hit:	100.00%	100	-	
bin max_neg	38	1	-	Covered
bin zero	50	1	-	Covered
bin max_pos	52	1	-	Covered
default bin misc	9870		-	Occurred
Coverpoint opcode_cp	100.00%	100	-	Covered
covered/total bins:	6	6	-	
missing/total bins:	0	6	-	
% Hit:	100.00%	100	-	
illegal_bin no_div	2696		-	Occurred
bin add_sub	4850	1	-	Covered
bin add_mult	4954	1	-	Covered
bin mult_sub	4824	1	-	Covered
bin add_then_sub	270	1	-	Covered
bin add_then_mult	268	1	-	Covered
bin mult_then_sub	294	1	-	Covered
Cross opcode_operand1	100.00%	100	-	Covered
covered/total bins:	6	6	-	
missing/total bins:	0	6	-	
% Hit:	100.00%	100	-	
Auto, Default and User Defined Bins:				
bin max_pos_add_or_sub	18	1	-	Covered
bin max_pos_add_or_mult	20	1	-	Covered
bin max_pos_mult_or_sub	18	1	-	Covered
bin max_neg_add_or_sub	22	1	-	Covered
bin max_neg_add_or_mult	26	1	-	Covered
bin max_neg_mult_or_sub	20	1	-	Covered
Cross opcode_operand2	100.00%	100	-	Covered
covered/total bins:	6	6	-	
missing/total bins:	0	6	-	
% Hit:	100.00%	100	-	
Auto, Default and User Defined Bins:				
bin max_pos_add_or_sub	28	1	-	Covered
bin max_pos_add_or_mult	14	1	-	Covered
bin max_pos_mult_or_sub	26	1	-	Covered
bin max_neg_add_or_sub	10	1	-	Covered
bin max_neg_add_or_mult	24	1	-	Covered

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```
    Cross opcode_operand2                          100.00%        100          -    Covered
        covered/total bins:                              6          6          -
        missing/total bins:                              0          6          -
        % Hit:                                     100.00%        100          -
        Auto, Default and User Defined Bins:
            bin max_pos_add_or_sub                      28          1          -    Covered
            bin max_pos_add_or_mult                     14          1          -    Covered
            bin max_pos_mult_or_sub                     26          1          -    Covered
            bin max_neg_add_or_sub                      10          1          -    Covered
            bin max_neg_add_or_mult                     24          1          -    Covered
            bin max_neg_mult_or_sub                     26          1          -    Covered
Statement Coverage:
    Enabled Coverage              Bins      Hits    Misses  Coverage
    ----------------              ----      ----    ------  --------
    Statements                       1         1         0   100.00%
```

code_cvr_alu - Notepad

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```
COVERGROUP COVERAGE:
----------------------------------------------------------------------------------------------------------
Covergroup                                                Metric       Goal       Bins    Status

----------------------------------------------------------------------------------------------------------
 TYPE /testing_pkg/Transaction/CovCode                   100.00%        100          -    Covered
    covered/total bins:                                       24         24          -
    missing/total bins:                                        0         24          -
    % Hit:                                                100.00%        100          -
    Coverpoint operand1_cp                                100.00%        100          -    Covered
        covered/total bins:                                    3          3          -
        missing/total bins:                                    0          3          -
        % Hit:                                            100.00%        100          -
    Coverpoint operand2_cp                                100.00%        100          -    Covered
        covered/total bins:                                    3          3          -
        missing/total bins:                                    0          3          -
        % Hit:                                            100.00%        100          -
    Coverpoint opcode_cp                                  100.00%        100          -    Covered
        covered/total bins:                                    6          6          -
        missing/total bins:                                    0          6          -
        % Hit:                                            100.00%        100          -
    Cross opcode_operand1                                 100.00%        100          -    Covered
        covered/total bins:                                    6          6          -
        missing/total bins:                                    0          6          -
        % Hit:                                            100.00%        100          -
    Cross opcode_operand2                                 100.00%        100          -    Covered
        covered/total bins:                                    6          6          -
        missing/total bins:                                    0          6          -
        % Hit:                                            100.00%        100          -
 Covergroup instance \/testing_pkg::Transaction::CovCode
                                                          100.00%        100          -    Covered
    covered/total bins:                                       24         24          -
    missing/total bins:                                        0         24          -
    % Hit:                                                100.00%        100          -
    Coverpoint operand1_cp                                100.00%        100          -    Covered
        covered/total bins:                                    3          3          -
        missing/total bins:                                    0          3          -
        % Hit:                                            100.00%        100          -
        bin max_neg                                           44          1          -    Covered
        bin zero                                              48          1          -    Covered
        bin max_pos                                           44          1          -    Covered
        default bin misc                                    9874                     -    Occurred
    Coverpoint operand2_cp                                100.00%        100          -    Covered
```

code_cvr_alu - Notepad

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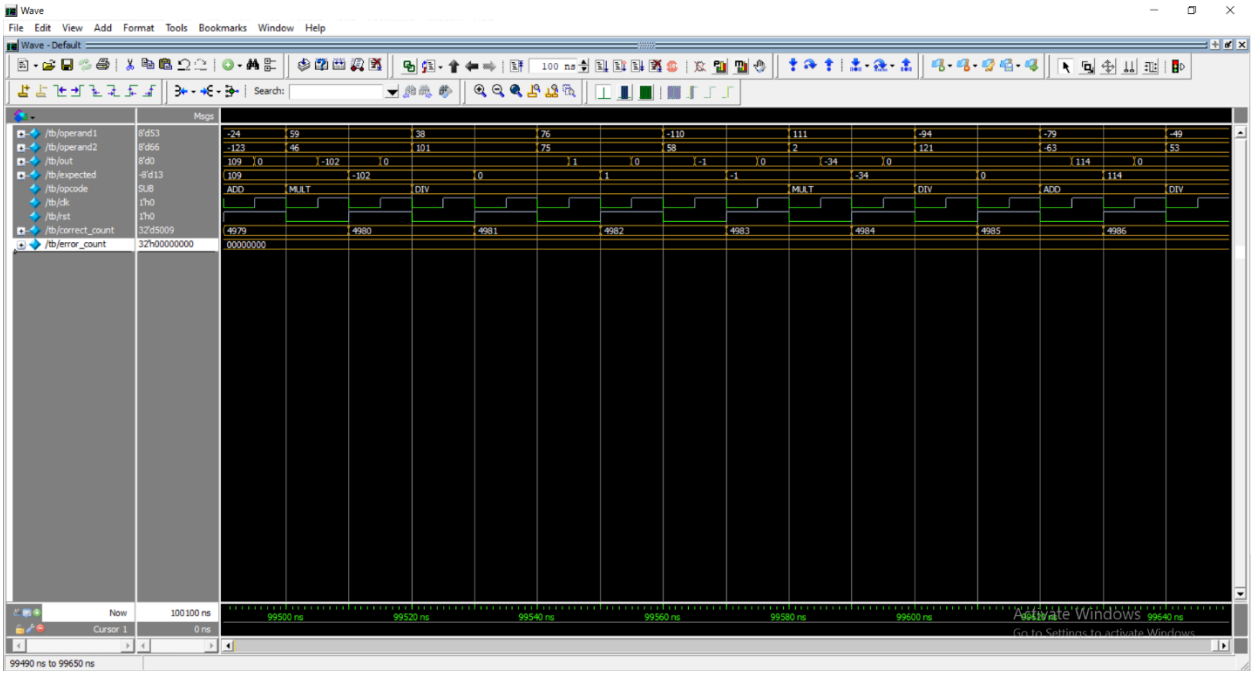
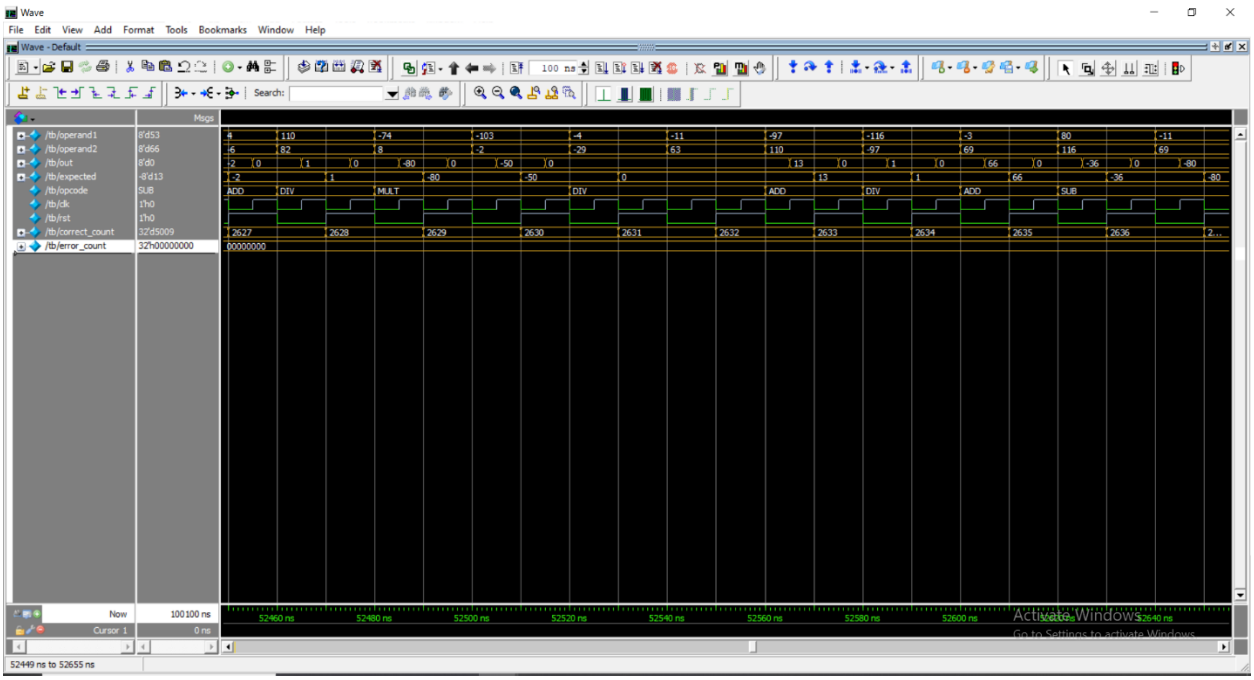
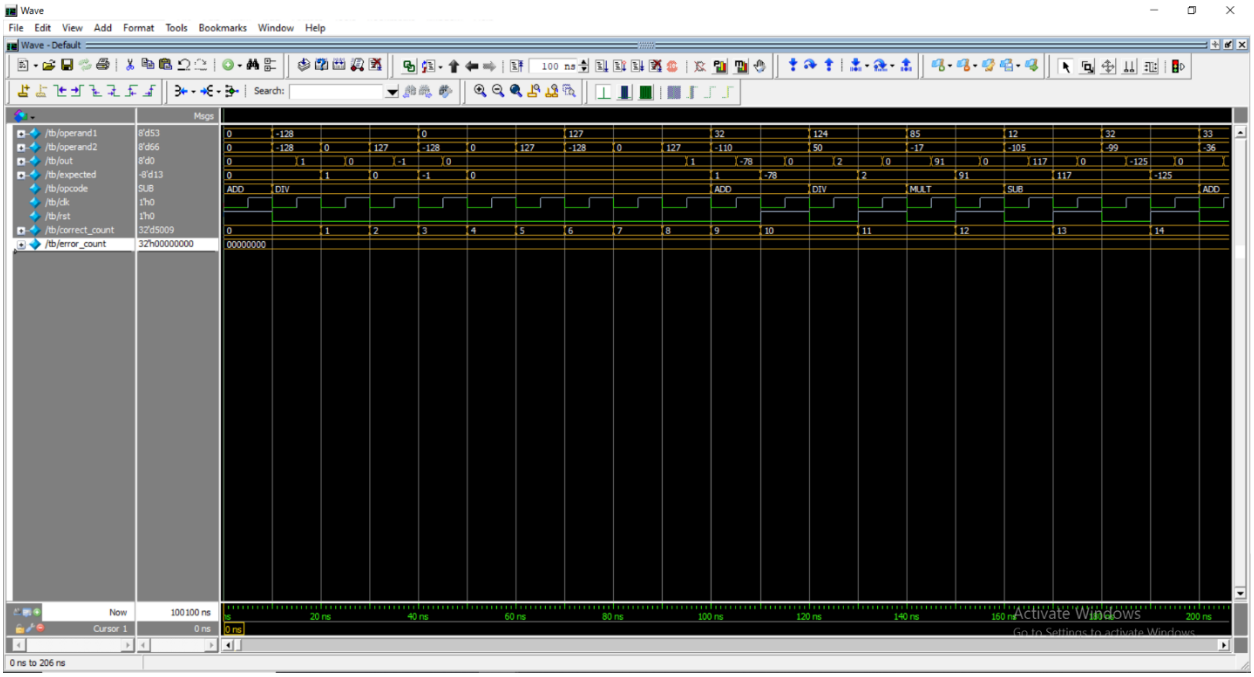
```
    Coverpoint operand2_cp                          100.00%        100          -    Covered
        covered/total bins:                               3          3          -
        missing/total bins:                               0          3          -
        % Hit:                                       100.00%        100          -
        bin max_neg                                      38          1          -    Covered
        bin zero                                         50          1          -    Covered
        bin max_pos                                      52          1          -    Covered
        default bin misc                               9870                     -    Occurred
    Coverpoint opcode_cp                            100.00%        100          -    Covered
        covered/total bins:                               6          6          -
        missing/total bins:                               0          6          -
        % Hit:                                       100.00%        100          -
        illegal_bin no_div                             2696                     -    Occurred
        bin add_sub                                    4850          1          -    Covered
        bin add_mult                                   4954          1          -    Covered
        bin mult_sub                                   4824          1          -    Covered
        bin add_then_sub                                270          1          -    Covered
        bin add_then_mult                               268          1          -    Covered
        bin mult_then_sub                               294          1          -    Covered
    Cross opcode_operand1                           100.00%        100          -    Covered
        covered/total bins:                               6          6          -
        missing/total bins:                               0          6          -
        % Hit:                                       100.00%        100          -
        Auto, Default and User Defined Bins:
            bin max_pos_add_or_sub                       18          1          -    Covered
            bin max_pos_add_or_mult                      20          1          -    Covered
            bin max_pos_mult_or_sub                      18          1          -    Covered
            bin max_neg_add_or_sub                       22          1          -    Covered
            bin max_neg_add_or_mult                      26          1          -    Covered
            bin max_neg_mult_or_sub                      20          1          -    Covered
    Cross opcode_operand2                           100.00%        100          -    Covered
        covered/total bins:                               6          6          -
        missing/total bins:                               0          6          -
        % Hit:                                       100.00%        100          -
        Auto, Default and User Defined Bins:
            bin max_pos_add_or_sub                       28          1          -    Covered
            bin max_pos_add_or_mult                      14          1          -    Covered
            bin max_pos_mult_or_sub                      26          1          -    Covered
            bin max_neg_add_or_sub                       10          1          -    Covered
            bin max_neg_add_or_mult                      24          1          -    Covered
            bin max_neg_mult_or_sub                      26          1          -    Covered

TOTAL COVERGROUP COVERAGE: 100.00%  COVERGROUP TYPES: 1
```

9. Clear and neat QuestaSim waveform snippets showing the functionality of the design


Wave
File Edit View Add Format Tools Bookmarks Window Help
Wave - Default
/tb/operand1
/tb/operand2
/tb/out
/tb/expected
/tb/opcode
/tb/clk
/tb/rst
/tb/correct_count
/tb/error_count
Now
Cursor 1
99490 ns to 99650 ns
Activate Windows
Go to Settings to activate Windows.


Wave
File Edit View Add Format Tools Bookmarks Window Help
Wave - Default
/tb/operand1
/tb/operand2
/tb/out
/tb/expected
/tb/opcode
/tb/clk
/tb/rst
/tb/correct_count
/tb/error_count
Now
Cursor 1
52449 ns to 52655 ns
Activate Windows
Go to Settings to activate Windows.


Wave
File Edit View Add Format Tools Bookmarks Window Help
Wave - Default
Search:
Msgs
/tb/operand1
/tb/operand2
/tb/out
/tb/expected
/tb/opcode
/tb/clk
/tb/rst
/tb/correct_count
/tb/error_count
8'd53
8'd66
8'd0
-8'd13
SUB
1'h0
1'h0
32'd5009
32'h00000000
Now
Cursor 1
100 100 ns
0 ns
0 ns to 206 ns
Activate Windows
Go to Settings to activate Windows.

Q2

1. Testbench code

```
import question_2_pkg::*;

module question_2_tb();
  parameter int WIDTH = question_2_pkg::WIDTH;

  // ============================
  // DUT signals
  // ============================
  logic clk;
  logic rst_n, load_n, up_down, ce;
  logic [WIDTH-1:0] data_load;
  logic [WIDTH-1:0] count_out;
  logic max_count, zero;

  // expected
  logic [WIDTH-1:0] count_out_expected;
  bit expected_max;
  bit expected_zero;

  // class object
  question_2_class question_2_object;

  // ============================
  // DUT instantiation
  // ============================
  counter #(.WIDTH(WIDTH)) dut (
    .clk(clk),
    .rst_n(rst_n),
    .load_n(load_n),
    .up_down(up_down),
    .ce(ce),
    .data_load(data_load),
    .count_out(count_out),
    .max_count(max_count),
    .zero(zero)
  );
```

```
38    // ============================
39    // Clock generation
40    // ============================
41    initial begin
42      clk = 0;
43      forever #5 clk = ~clk; // 10ns period
44    end
45
46    // ============================
47    // Tasks
48    // ============================
49    task assert_reset();
50      begin
51        rst_n = 0;
52        @(posedge clk);
53        rst_n = 1;
54        @(posedge clk);
55      end
56    endtask
57
58    task golden_model (
59      input logic [WIDTH-1:0] prev_count,
60      input bit load_n_in, ce_in, up_down_in,
61      input logic [WIDTH-1:0] data_load_in,
62      output logic [WIDTH-1:0] expected,
63      output bit expected_max_o,
64      output bit expected_zero_o
65    );
66      if (!load_n_in)
67        expected = data_load_in;
68      else if (ce_in)
69        expected = up_down_in ? prev_count + 1 : prev_count - 1;
70      else
71        expected = prev_count;
72
```

```
    expected_max_o  = (expected == {WIDTH{1'b1}});
    expected_zero_o = (expected == 0);
  endtask

  task check_result(
    input logic [WIDTH-1:0] expected,
    input bit expected_max_i,
    input bit expected_zero_i
  );
    if (count_out !== expected)
      $display("ERROR @%0t: count_out mismatch. got=%0d expected=%0d",
               $time, count_out, expected);
    if (max_count !== expected_max_i)
      $display("ERROR @%0t: max_count mismatch. got=%b expected=%b",
               $time, max_count, expected_max_i);
    if (zero !== expected_zero_i)
      $display("ERROR @%0t: zero mismatch. got=%b expected=%b",
               $time, zero, expected_zero_i);
    if ((count_out === expected) &&
        (max_count === expected_max_i) &&
        (zero === expected_zero_i))
      $display("PASS @%0t: count_out=%0d, max=%b, zero=%b",
               $time, count_out, max_count, zero);
  endtask

```

```
98    // ============================
99    // Stimulus
100   // ============================
101   initial begin
102     // init signals
103     rst_n     = 1'b1;
104     load_n    = 1'b1;
105     ce        = 1'b0;
106     up_down   = 1'b1;
107     data_load = '0;
108
109     count_out_expected = '0;
110     expected_max  = 1'b0;
111     expected_zero = 1'b1;
112
113     // create class object
114     question_2_object = new();
115
116     // apply reset
117     assert_reset();
118
119     // run randomized tests
120     repeat (40000) begin
121
122       // save previous count before DUT update
123       static logic [WIDTH-1:0] prev_count = count_out;
124
125       // randomize fields in class
126       assert(question_2_object.randomize())
127         else $fatal("Randomization failed!");
128
```

```
129         // drive DUT inputs
130         load_n    = question_2_object.load_n_class;
131         ce        = question_2_object.ce_class;
132         up_down   = question_2_object.up_down_class;
133         data_load = question_2_object.data_load_class;
134
135         // wait for DUT update
136         @(posedge clk);
137
138         // compute expected output
139         golden_model(prev_count, load_n, ce, up_down, data_load,
140                      count_out_expected, expected_max, expected_zero);
141
142         // sample coverage
143         question_2_object.sampled_count = count_out;
144         question_2_object.counter_cov.sample();
145
146         // check result
147         check_result(count_out_expected, expected_max, expected_zero);
148       end
149
150       $display("=== TEST COMPLETE ===");
151       $finish;
152     end
153
154   endmodule
```

2. Package code

```
package question_2_pkg;
  parameter int WIDTH = 4;

  class question_2_class;
    // random control fields
    rand logic rst_n_class, load_n_class, up_down_class, ce_class;
    rand logic [WIDTH-1:0] data_load_class;

    // variable that TB will fill with DUT count for coverage sampling
    logic [WIDTH-1:0] sampled_count;

    // Constraints / distributions
    constraint distribution {
      rst_n_class  dist {1 := 80, 0 := 20};
      load_n_class dist {0 := 70, 1 := 30};
      ce_class     dist {1 := 70, 0 := 30};
    }

    // Covergroup defined inside class (no event control here)
    covergroup counter_cov;
      // 1. load-data when load asserted and reset deasserted
      load_data_cp: coverpoint data_load_class
        iff (load_n_class == 0 && rst_n_class == 1);

      // 2. count-up values (sampled_count) when rst=1, ce=1, up_down=1
      count_up_cp: coverpoint sampled_count
        iff (rst_n_class == 1 && ce_class == 1 && up_down_class == 1);

      // 3. overflow detection (max -> 0) as transition bin
      count_up_overflow: coverpoint sampled_count
        iff (rst_n_class == 1 && ce_class == 1 && up_down_class == 1) {
        bins overflow = ({WIDTH{1'b1}} => 0);
      }

```

```
34
35        // 4. count-down values when rst=1, ce=1, up_down=0
36        count_down_cp: coverpoint sampled_count
37          iff (rst_n_class == 1 && ce_class == 1 && up_down_class == 0);
38
39        // 5. underflow detection (0 -> max) as transition bin
40        count_down_underflow: coverpoint sampled_count
41          iff (rst_n_class == 1 && ce_class == 1 && up_down_class == 0) {
42          bins underflow = (0 => {WIDTH{1'b1}});
43        }
44      endgroup
45
46      function new();
47        counter_cov = new();
48      endfunction
49
50    endclass
51
52  endpackage
53
```

3. Design code

```
1  ////////////////////////////////////////////////////////////////////////////////
2  // Author: Kareem Waseem
3  // Course: Digital Verification using SV & UVM
4  //
5  // Description: Counter Design
6  //
7  ////////////////////////////////////////////////////////////////////////////////
8  module counter (clk ,rst_n, load_n, up_down, ce, data_load, count_out, max_count, zero);
9  parameter WIDTH = 4;
10 input clk;
11 input rst_n;
12 input load_n;
13 input up_down;
14 input ce;
15 input [WIDTH-1:0] data_load;
16 output reg [WIDTH-1:0] count_out;
17 output max_count;
18 output zero;
19
20 always @(posedge clk) begin
21     if (!rst_n)
22         count_out <= 0;
23     else if (!load_n)
24         count_out <= data_load;
25     else if (ce)
26         if (up_down)
27             count_out <= count_out + 1;
28         else
29             count_out <= count_out - 1;
30 end
31
32 assign max_count = (count_out == {WIDTH{1'b1}})? 1:0;
33 assign zero = (count_out == 0)? 1:0;
34
35 endmodule
```

4. Report any bugs detected in the design and fix them

5. Snippet to your verification plan document

Label	Design requirement description	Stimulus generation	Functional coverage	Functionality check
COUNTER_RESET	When reset is asserted, count_out = 0.	Directed at start	Coverpoint implicitly skipped since reset is 0.	Task compares DUT output with expected=0.
COUNTER_LOAD	When load is asserted, count_out = data_load.	Random + Directed corners	load_data_cp coverpoint samples data_load when load_n=0, rst_n=1.	Task golden_model ensures DUT output matches data_load.
COUNTER_COUNT_UP	When up_down=1, enable=1, DUT counts upward.	Random + Directed	count_up_cp auto-bins all values of count_out.	Task golden_model + check_result.
COUNTER_OVERFLOW	When counting up, after max value counter wraps around to 0.	Directed (set to max, ce=1)	count_up_overflow transition bin checks {max → 0}.	Checker compares expected wraparound.
COUNTER_COUNT_DOWN	When up_down=0, enable=1, DUT counts downward.	Random + Directed	count_down_cp auto-bins all values of count_out.	Task golden_model + check_result.
COUNTER_UNDERFLOW	When counting down, after 0 the counter wraps to max.	Directed (set to 0, ce=1)	count_down_underflow transition bin checks {0 → max}.	Checker compares expected wraparound.
COUNTER_ENABLE_HOLD	If enable=0, count_out must hold its previous value.	Random	Can extend coverage with cross of ce=0 & count_out.	Task ensures DUT output equals previous value.

6. Do file

```
vlib work
vlog counter.v package.svh question_2_tb.svh  +cover -covercells
vsim -voptargs=+acc work.question_2_tb -cover
add wave *
coverage save question_2_tb.ucdb -onexit
run -all
```

7. Code Coverage report snippets (Questions 1, 2, and 3 only)

code_cvr_alu - Notepad

File Edit Format View Help

```
Coverage Report by instance with details

================================================================================
=== Instance: /question_2_tb/dut
=== Design Unit: work.counter
================================================================================
Branch Coverage:
    Enabled Coverage              Bins      Hits    Misses  Coverage
    ----------------              ----      ----    ------  --------
    Branches                        10        10         0   100.00%

================================Branch Details================================

Branch Coverage for instance /question_2_tb/dut

    Line         Item                      Count     Source
    ----         ----                      -----     ------
  File counter.v
------------------------------------IF Branch------------------------------------
    21                                     39966     Count coming in to IF
    21              1                          1         if (!rst_n)

    23              1                      27873         else if (!load_n)

    25              1                       8483         else if (ce)

                                            3609     All False Count
Branch totals: 4 hits of 4 branches = 100.00%

------------------------------------IF Branch------------------------------------
    26                                      8483     Count coming in to IF
    26              1                       4274             if (up_down)

    28              1                       4209             else

Branch totals: 2 hits of 2 branches = 100.00%

------------------------------------IF Branch------------------------------------
    32                                     34590     Count coming in to IF
    32              1                       2163     assign max_count = (count_out == {WIDTH{1'b1}})? 1:0;

    32              2                      32427     assign max_count = (count_out == {WIDTH{1'b1}})? 1:0;
```

code_cvr_alu - Notepad

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```
Branch totals: 2 hits of 2 branches = 100.00%

------------------------------------IF Branch------------------------------------
    33                                    34590     Count coming in to IF
    33              1                      2182     assign zero = (count_out == 0)? 1:0;

    33              2                     32408     assign zero = (count_out == 0)? 1:0;

Branch totals: 2 hits of 2 branches = 100.00%


Condition Coverage:
    Enabled Coverage              Bins   Covered    Misses  Coverage
    ----------------              ----      ----    ------  --------
    Conditions                       2         2         0   100.00%

================================Condition Details================================

Condition Coverage for instance /question_2_tb/dut --

  File counter.v
----------------Focused Condition View-------------------
Line       32 Item    1  (count_out == {4{{1}}})
Condition totals: 1 of 1 input term covered = 100.00%

               Input Term   Covered  Reason for no coverage   Hint
              -----------  --------  -----------------------  --------------
  (count_out == {4{{1}}})         Y

     Rows:       Hits  FEC Target                Non-masking condition(s)
 ---------  ---------  --------------------      -------------------------
  Row   1:          1  (count_out == {4{{1}}})_0  -
  Row   2:          1  (count_out == {4{{1}}})_1  -

----------------Focused Condition View-------------------
Line       33 Item    1  (count_out == 0)
Condition totals: 1 of 1 input term covered = 100.00%

        Input Term   Covered  Reason for no coverage   Hint
       -----------  --------  -----------------------  --------------
  (count_out == 0)         Y
```

code_cvr_alu - Notepad

File Edit Format View Help

```
     Rows:       Hits  FEC Target            Non-masking condition(s)
 ---------  ---------  --------------------  -------------------------
  Row   1:          1  (count_out == 0)_0    -
  Row   2:          1  (count_out == 0)_1    -


Statement Coverage:
    Enabled Coverage              Bins      Hits    Misses  Coverage
    ----------------              ----      ----    ------  --------
    Statements                       7         7         0   100.00%

================================Statement Details================================

Statement Coverage for instance /question_2_tb/dut --

    Line         Item                     Count     Source
    ----         ----                     -----     ------
  File counter.v
    8                                               module counter (clk ,rst_n, load_n, up_down, ce, data_load, count_out, max_count, zero);

    9                                               parameter WIDTH = 4;

    10                                              input clk;

    11                                              input rst_n;

    12                                              input load_n;

    13                                              input up_down;

    14                                              input ce;

    15                                              input [WIDTH-1:0] data_load;

    16                                              output reg [WIDTH-1:0] count_out;

    17                                              output max_count;

    18                                              output zero;

    19
```

code_cvr_alu - Notepad

File Edit Format View Help

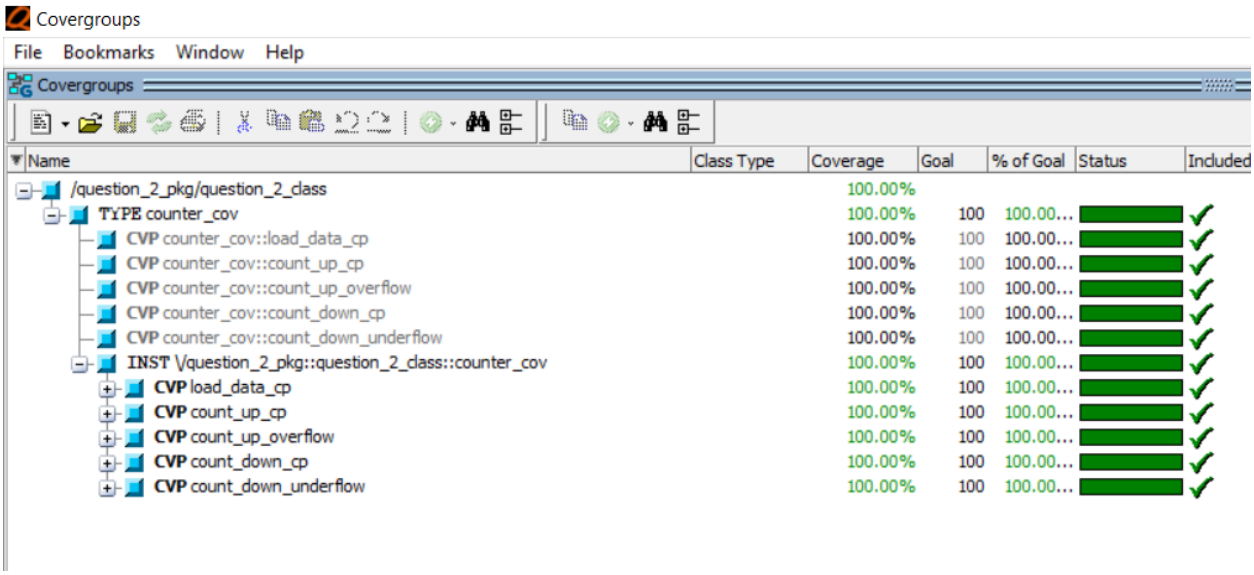
```
    20              1                      39966     always @(posedge clk) begin
    21                                                    if (!rst_n)
    22              1                          1             count_out <= 0;
    23                                                    else if (!load_n)
    24              1                      27873             count_out <= data_load;
    25                                                    else if (ce)
    26                                                        if (up_down)
    27              1                       4274                count_out <= count_out + 1;
    28                                                        else
    29              1                       4209                count_out <= count_out - 1;
    30                                                end
    31
    32              1                      34591     assign max_count = (count_out == {WIDTH{1'b1}})? 1:0;
    33              1                      34591     assign zero = (count_out == 0)? 1:0;


Toggle Coverage:
    Enabled Coverage              Bins      Hits    Misses  Coverage
    ----------------              ----      ----    ------  --------
    Toggles                         28        28         0   100.00%

================================Toggle Details================================

Toggle Coverage for instance /question_2_tb/dut --

                                          Node      1H->0L      0L->1H  "Coverage"
                                          ----------------------------------------
                                            ce           1           1      100.00
                                           clk           1           1      100.00
```

code_cvr_alu - Notepad

File Edit Format View Help

```
    Toggles                          28        28         0   100.00%

================================Toggle Details================================

Toggle Coverage for instance /question_2_tb/dut --

                                          Node      1H->0L      0L->1H  "Coverage"
                                          ----------------------------------------
                                            ce           1           1      100.00
                                           clk           1           1      100.00
                                  count_out[3-0]         1           1      100.00
                                  data_load[0-3]         1           1      100.00
                                          load_n         1           1      100.00
                                       max_count         1           1      100.00
                                         up_down         1           1      100.00
                                            zero         1           1      100.00

Total Node Count     =         14
Toggled Node Count   =         14
Untoggled Node Count =          0

Toggle Coverage      =     100.00% (28 of 28 bins)

================================================================================
```

8. Functional Coverage report snippets (Questions 1, 2, and 3 only)


Covergroups
File Bookmarks Window Help
Covergroups
Name
Class Type
Coverage
Goal
% of Goal
Status
Included
/question_2_pkg/question_2_class 100.00%
TYPE counter_cov 100.00% 100 100.00...
CVP counter_cov::load_data_cp 100.00% 100 100.00...
CVP counter_cov::count_up_cp 100.00% 100 100.00...
CVP counter_cov::count_up_overflow 100.00% 100 100.00...
CVP counter_cov::count_down_cp 100.00% 100 100.00...
CVP counter_cov::count_down_underflow 100.00% 100 100.00...
INST \/question_2_pkg::question_2_class::counter_cov 100.00% 100 100.00...
CVP load_data_cp 100.00% 100 100.00...
CVP count_up_cp 100.00% 100 100.00...
CVP count_up_overflow 100.00% 100 100.00...
CVP count_down_cp 100.00% 100 100.00...
CVP count_down_underflow 100.00% 100 100.00...

code_cvr_alu - Notepad

File Edit Format View Help

```
Covergroup Coverage:
    Covergroups                     1        na        na   100.00%
        Coverpoints/Crosses         5        na        na        na
            Covergroup Bins        50        50         0   100.00%
------------------------------------------------------------------------------------------------------------------------
Covergroup                                             Metric       Goal       Bins    Status

------------------------------------------------------------------------------------------------------------------------
 TYPE /question_2_pkg/question_2_class/counter_cov    100.00%        100          -    Covered
    covered/total bins:                                    50         50          -
    missing/total bins:                                     0         50          -
    % Hit:                                             100.00%        100          -
    Coverpoint load_data_cp                            100.00%        100          -    Covered
        covered/total bins:                                16         16          -
        missing/total bins:                                 0         16          -
        % Hit:                                         100.00%        100          -
    Coverpoint count_up_cp                             100.00%        100          -    Covered
        covered/total bins:                                16         16          -
        missing/total bins:                                 0         16          -
        % Hit:                                         100.00%        100          -
    Coverpoint count_up_overflow                       100.00%        100          -    Covered
        covered/total bins:                                 1          1          -
        missing/total bins:                                 0          1          -
        % Hit:                                         100.00%        100          -
    Coverpoint count_down_cp                           100.00%        100          -    Covered
        covered/total bins:                                16         16          -
        missing/total bins:                                 0         16          -
        % Hit:                                         100.00%        100          -
    Coverpoint count_down_underflow                    100.00%        100          -    Covered
        covered/total bins:                                 1          1          -
        missing/total bins:                                 0          1          -
        % Hit:                                         100.00%        100          -
 Covergroup instance \/question_2_pkg::question_2_class::counter_cov
                                                       100.00%        100          -    Covered
    covered/total bins:                                    50         50          -
    missing/total bins:                                     0         50          -
    % Hit:                                             100.00%        100          -
    Coverpoint load_data_cp                            100.00%        100          -    Covered
        covered/total bins:                                16         16          -
        missing/total bins:                                 0         16          -
        % Hit:                                         100.00%        100          -
        bin auto[0]                                       1447          1          -    Covered
```

code_cvr_alu - Notepad

File Edit Format View Help

```
        bin auto[0]                          1447          1          -    Covered
        bin auto[1]                          1408          1          -    Covered
        bin auto[2]                          1390          1          -    Covered
        bin auto[3]                          1391          1          -    Covered
        bin auto[4]                          1329          1          -    Covered
        bin auto[5]                          1322          1          -    Covered
        bin auto[6]                          1328          1          -    Covered
        bin auto[7]                          1413          1          -    Covered
        bin auto[8]                          1417          1          -    Covered
        bin auto[9]                          1377          1          -    Covered
        bin auto[10]                         1396          1          -    Covered
        bin auto[11]                         1308          1          -    Covered
        bin auto[12]                         1411          1          -    Covered
        bin auto[13]                         1429          1          -    Covered
        bin auto[14]                         1409          1          -    Covered
        bin auto[15]                         1375          1          -    Covered
    Coverpoint count_up_cp                 100.00%       100          -    Covered
        covered/total bins:                     16        16          -
        missing/total bins:                      0        16          -
        % Hit:                             100.00%       100          -
        bin auto[0]                            695         1          -    Covered
        bin auto[1]                            701         1          -    Covered
        bin auto[2]                            687         1          -    Covered
        bin auto[3]                            718         1          -    Covered
        bin auto[4]                            640         1          -    Covered
        bin auto[5]                            644         1          -    Covered
        bin auto[6]                            700         1          -    Covered
        bin auto[7]                            714         1          -    Covered
        bin auto[8]                            696         1          -    Covered
        bin auto[9]                            690         1          -    Covered
        bin auto[10]                           718         1          -    Covered
        bin auto[11]                           682         1          -    Covered
        bin auto[12]                           699         1          -    Covered
        bin auto[13]                           727         1          -    Covered
        bin auto[14]                           749         1          -    Covered
        bin auto[15]                           686         1          -    Covered
    Coverpoint count_up_overflow           100.00%       100          -    Covered
        covered/total bins:                      1         1          -
        missing/total bins:                      0         1          -
        % Hit:                             100.00%       100          -
        bin overflow                            94         1          -    Covered
    Coverpoint count_down_cp               100.00%       100          -    Covered
        covered/total bins:                     16        16          -
```

code_cvr_alu - Notepad

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```
        covered/total bins:                                16         16          -
        missing/total bins:                                 0         16          -
        % Hit:                                         100.00%        100          -
        bin auto[0]                                       732          1          -    Covered
        bin auto[1]                                       718          1          -    Covered
        bin auto[2]                                       706          1          -    Covered
        bin auto[3]                                       678          1          -    Covered
        bin auto[4]                                       683          1          -    Covered
        bin auto[5]                                       645          1          -    Covered
        bin auto[6]                                       648          1          -    Covered
        bin auto[7]                                       671          1          -    Covered
        bin auto[8]                                       708          1          -    Covered
        bin auto[9]                                       680          1          -    Covered
        bin auto[10]                                      698          1          -    Covered
        bin auto[11]                                      659          1          -    Covered
        bin auto[12]                                      736          1          -    Covered
        bin auto[13]                                      737          1          -    Covered
        bin auto[14]                                      661          1          -    Covered
        bin auto[15]                                      710          1          -    Covered
    Coverpoint count_down_underflow                    100.00%        100          -    Covered
        covered/total bins:                                 1          1          -
        missing/total bins:                                 0          1          -
        % Hit:                                         100.00%        100          -
        bin underflow                                      97          1          -    Covered
Statement Coverage:
    Enabled Coverage              Bins      Hits    Misses  Coverage
    ----------------              ----      ----    ------  --------
    Statements                       1         1         0   100.00%
```

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```
COVERGROUP COVERAGE:
----------------------------------------------------------------------------------------------------
Covergroup                                              Metric       Goal       Bins    Status

----------------------------------------------------------------------------------------------------
 TYPE /question_2_pkg/question_2_class/counter_cov    100.00%        100          -    Covered
    covered/total bins:                                    50         50          -
    missing/total bins:                                     0         50          -
    % Hit:                                             100.00%        100          -
    Coverpoint load_data_cp                            100.00%        100          -    Covered
        covered/total bins:                                16         16          -
        missing/total bins:                                 0         16          -
        % Hit:                                         100.00%        100          -
    Coverpoint count_up_cp                             100.00%        100          -    Covered
        covered/total bins:                                16         16          -
        missing/total bins:                                 0         16          -
        % Hit:                                         100.00%        100          -
    Coverpoint count_up_overflow                       100.00%        100          -    Covered
        covered/total bins:                                 1          1          -
        missing/total bins:                                 0          1          -
        % Hit:                                         100.00%        100          -
    Coverpoint count_down_cp                           100.00%        100          -    Covered
        covered/total bins:                                16         16          -
        missing/total bins:                                 0         16          -
        % Hit:                                         100.00%        100          -
    Coverpoint count_down_underflow                    100.00%        100          -    Covered
        covered/total bins:                                 1          1          -
        missing/total bins:                                 0          1          -
        % Hit:                                         100.00%        100          -
 Covergroup instance \/question_2_pkg::question_2_class::counter_cov
                                                       100.00%        100          -    Covered
    covered/total bins:                                    50         50          -
    missing/total bins:                                     0         50          -
    % Hit:                                             100.00%        100          -
    Coverpoint load_data_cp                            100.00%        100          -    Covered
        covered/total bins:                                16         16          -
        missing/total bins:                                 0         16          -
        % Hit:                                         100.00%        100          -
        bin auto[0]                                       1447          1          -    Covered
        bin auto[1]                                       1408          1          -    Covered
        bin auto[2]                                       1390          1          -    Covered
        bin auto[3]                                       1391          1          -    Covered
        bin auto[4]                                       1329          1          -    Covered
```

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bin auto[4]	1329	1	-	Covered
bin auto[5]	1322	1	-	Covered
bin auto[6]	1328	1	-	Covered
bin auto[7]	1413	1	-	Covered
bin auto[8]	1417	1	-	Covered
bin auto[9]	1377	1	-	Covered
bin auto[10]	1396	1	-	Covered
bin auto[11]	1308	1	-	Covered
bin auto[12]	1411	1	-	Covered
bin auto[13]	1429	1	-	Covered
bin auto[14]	1409	1	-	Covered
bin auto[15]	1375	1	-	Covered
Coverpoint count_up_cp	100.00%	100	-	Covered
covered/total bins:	16	16	-	
missing/total bins:	0	16	-	
% Hit:	100.00%	100	-	
bin auto[0]	695	1	-	Covered
bin auto[1]	701	1	-	Covered
bin auto[2]	687	1	-	Covered
bin auto[3]	718	1	-	Covered
bin auto[4]	640	1	-	Covered
bin auto[5]	644	1	-	Covered
bin auto[6]	700	1	-	Covered
bin auto[7]	714	1	-	Covered
bin auto[8]	696	1	-	Covered
bin auto[9]	690	1	-	Covered
bin auto[10]	718	1	-	Covered
bin auto[11]	682	1	-	Covered
bin auto[12]	699	1	-	Covered
bin auto[13]	727	1	-	Covered
bin auto[14]	749	1	-	Covered
bin auto[15]	686	1	-	Covered
Coverpoint count_up_overflow	100.00%	100	-	Covered
covered/total bins:	1	1	-	
missing/total bins:	0	1	-	
% Hit:	100.00%	100	-	
bin overflow	94	1	-	Covered
Coverpoint count_down_cp	100.00%	100	-	Covered
covered/total bins:	16	16	-	
missing/total bins:	0	16	-	
% Hit:	100.00%	100	-	
bin auto[0]	732	1	-	Covered
bin auto[1]	718	1	-	Covered

code_cvr_alu - Notepad

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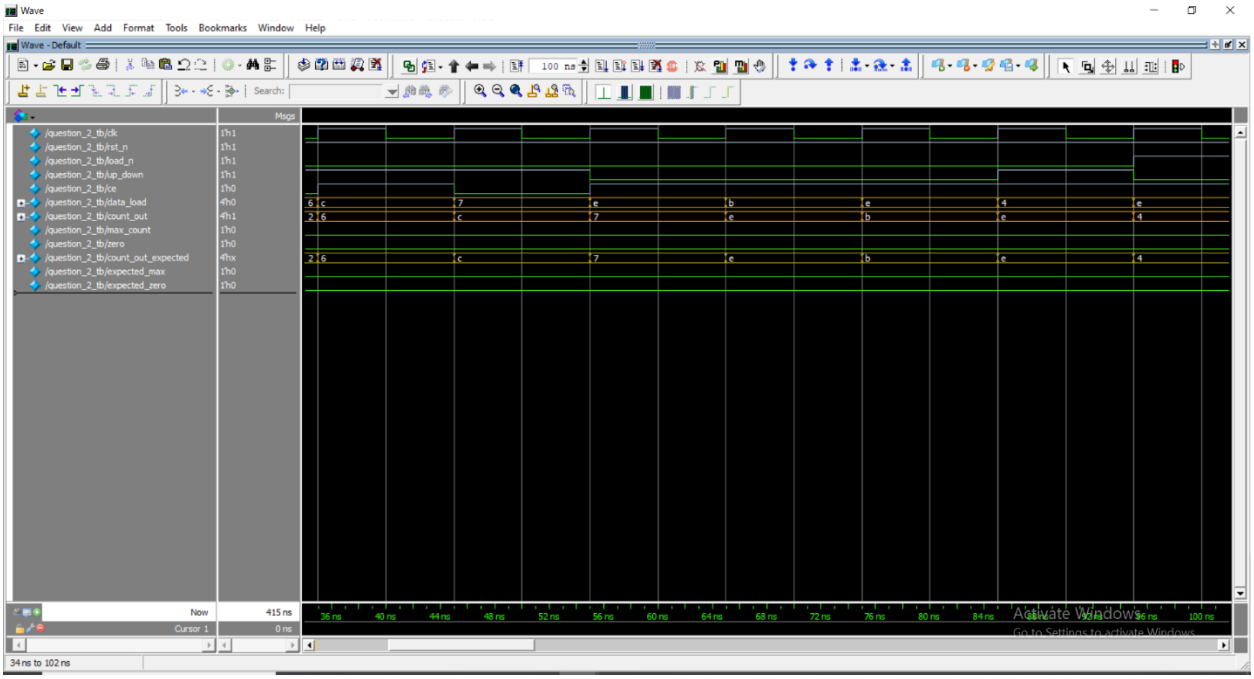
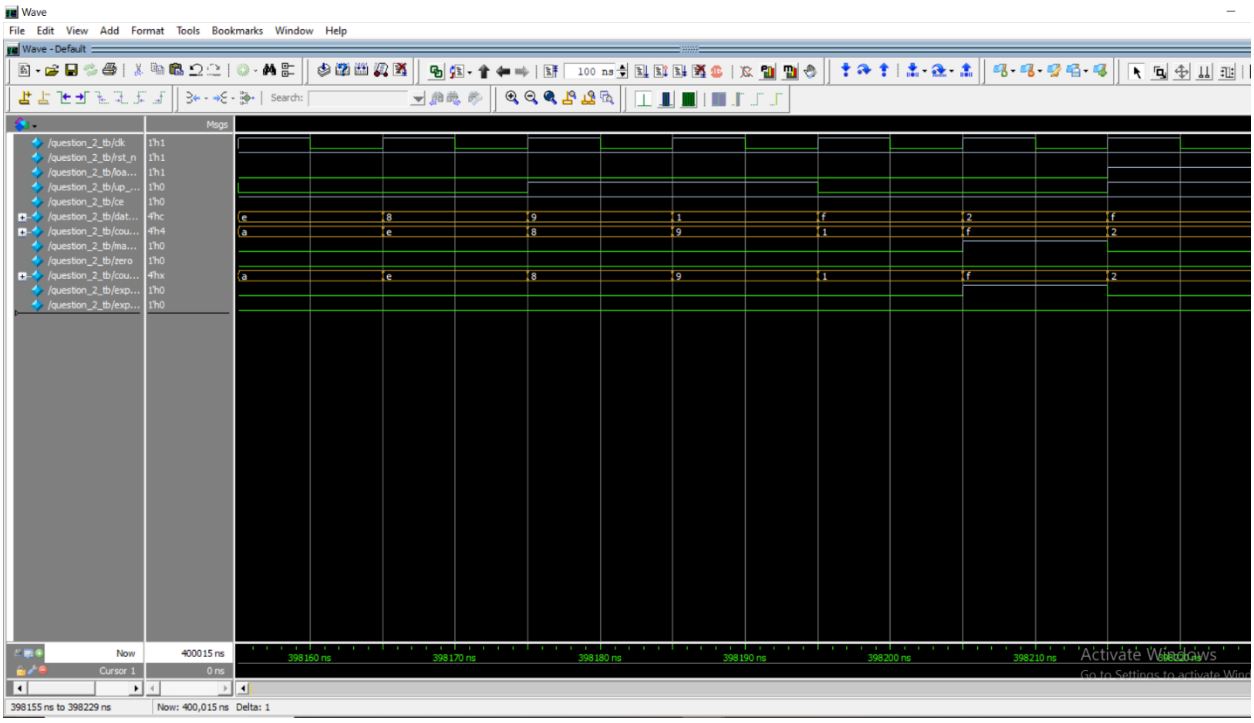
```
        covered/total bins:                      1         1          -
        missing/total bins:                      0         1          -
        % Hit:                             100.00%       100          -
        bin overflow                            94         1          -    Covered
    Coverpoint count_down_cp                100.00%       100          -    Covered
        covered/total bins:                     16        16          -
        missing/total bins:                      0        16          -
        % Hit:                             100.00%       100          -
        bin auto[0]                            732         1          -    Covered
        bin auto[1]                            718         1          -    Covered
        bin auto[2]                            706         1          -    Covered
        bin auto[3]                            678         1          -    Covered
        bin auto[4]                            683         1          -    Covered
        bin auto[5]                            645         1          -    Covered
        bin auto[6]                            648         1          -    Covered
        bin auto[7]                            671         1          -    Covered
        bin auto[8]                            708         1          -    Covered
        bin auto[9]                            680         1          -    Covered
        bin auto[10]                           698         1          -    Covered
        bin auto[11]                           659         1          -    Covered
        bin auto[12]                           736         1          -    Covered
        bin auto[13]                           737         1          -    Covered
        bin auto[14]                           661         1          -    Covered
        bin auto[15]                           710         1          -    Covered
    Coverpoint count_down_underflow         100.00%       100          -    Covered
        covered/total bins:                      1         1          -
        missing/total bins:                      0         1          -
        % Hit:                             100.00%       100          -
        bin underflow                           97         1          -    Covered

TOTAL COVERGROUP COVERAGE: 100.00%  COVERGROUP TYPES: 1
```

9. Clear and neat QuestaSim waveform snippets showing the functionality of the design





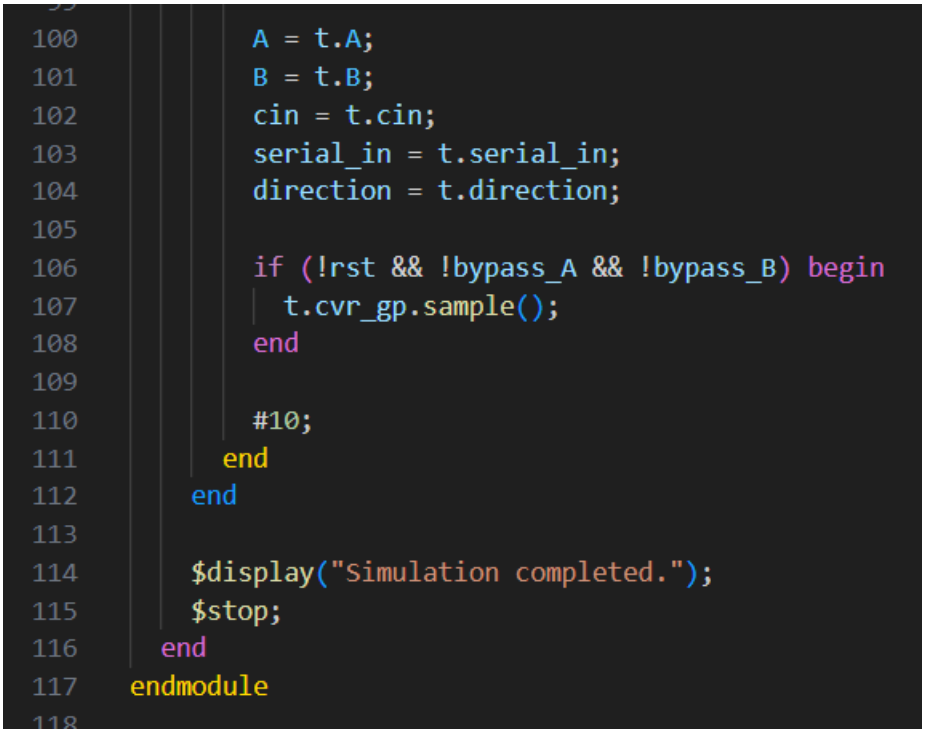
Q3

1. Testbench code

```
import alsu_pkg::*;

module question_3_tb;

  // DUT Signals
  logic clk, rst;
  logic signed [2:0] A, B;
  logic cin, serial_in;
  logic red_op_A, red_op_B;
  logic bypass_A, bypass_B;
  logic direction;
  opcode_e opcode;
  logic signed [5:0] result;

  // Transaction object
  alsu_txn t;

  // Instantiate DUT
  ALSU DUT (
    .clk(clk),
    .rst(rst),
    .A(A),
    .B(B),
    .cin(cin),
    .serial_in(serial_in),
    .red_op_A(red_op_A),
    .red_op_B(red_op_B),
    .bypass_A(bypass_A),
    .bypass_B(bypass_B),
    .direction(direction),
    .opcode(opcode),
    .out(result)
  );

```

```
36    // Clock generation
37    always #5 clk = ~clk;
38
39    initial begin
40      clk = 0;
41      rst = 1;
42      #20 rst = 0;
43
44      t = new();
45
46      // ---------- First Loop ----------
47      t.en_c8 = 0; // disable constraint #8
48      repeat (1000) begin
49        if (!t.randomize()) begin
50          $display("Randomization failed in first loop!");
51        end
52
53        // Drive DUT inputs
54        A = t.A;
55        B = t.B;
56        cin = t.cin;
57        serial_in = t.serial_in;
58        red_op_A = t.red_op_A;
59        red_op_B = t.red_op_B;
60        bypass_A = t.bypass_A;
61        bypass_B = t.bypass_B;
62        direction = t.direction;
63        opcode = t.opcode;
64
65        // Coverage sampling condition
66        if (!rst && !bypass_A && !bypass_B) begin
67          t.cvr_gp.sample();
68        end
69
```

```
70          #10;
71        end
72
73        // ---------- Second Loop ----------
74        $display("Starting second loop for opcode sequence...");
75        rst = 0;
76        bypass_A = 0;
77        bypass_B = 0;
78        red_op_A = 0;
79        red_op_B = 0;
80
81        t.en_c1 = 0;
82        t.en_c2 = 0;
83        t.en_c3 = 0;
84        t.en_c8 = 1; // enable constraint #8
85
86        // Generate unique opcode sequence
87        if (!t.randomize()) begin
88          $display("Randomization failed for opcode sequence!");
89        end
90
91        // Iterate through the unique opcode sequence
92        foreach (t.opcode_seq[i]) begin
93          opcode = t.opcode_seq[i];
94
95          repeat (6) begin
96            if (!t.randomize()) begin
97              $display("Randomization failed inside second loop!");
98            end
99
100           A = t.A;
101           B = t.B;
102           cin = t.cin;
```



```
100             A = t.A;
101             B = t.B;
102             cin = t.cin;
103             serial_in = t.serial_in;
104             direction = t.direction;
105
106             if (!rst && !bypass_A && !bypass_B) begin
107               t.cvr_gp.sample();
108             end
109
110             #10;
111           end
112         end
113
114         $display("Simulation completed.");
115         $stop;
116       end
117     endmodule
```

2. Package code

```
package alsu_pkg;

  // -------- ENUM for opcodes --------
  typedef enum logic [2:0] {
    OR_OP     = 3'h0,
    XOR_OP    = 3'h1,
    ADD_OP    = 3'h2,
    MUL_OP    = 3'h3,
    SHIFT_OP  = 3'h4,
    ROT_OP    = 3'h5,
    INVALID_6 = 3'h6,
    INVALID_7 = 3'h7
  } opcode_e;

  // -------- Transaction Class --------
  class alsu_txn;

    // Constants
    localparam signed [2:0] MAXPOS = 3;
    localparam signed [2:0] MAXNEG = -4;
    localparam signed [2:0] ZERO   = 0;

    // Randomizable Inputs
    rand logic clk, rst, cin, serial_in;
    rand logic red_op_A, red_op_B;
    rand logic bypass_A, bypass_B, direction;
    rand opcode_e opcode;
    rand logic signed [2:0] A, B;

    // Fixed array for constraint #8
    rand opcode_e opcode_seq[6];

    // Constraint enable flags
    bit en_c1, en_c2, en_c3, en_c4, en_c5, en_c6, en_c7, en_c8;

```

```
36    // -------- Covergroup --------
37    covergroup cvr_gp @(posedge clk);
38      option.per_instance = 1;
39
40      // A Coverpoint
41      coverpoint A {
42        bins A_data_0       = {0};
43        bins A_data_max     = {MAXPOS};
44        bins A_data_min     = {MAXNEG};
45        bins A_data_default = default;
46      }
47
48      // B Coverpoint
49      coverpoint B {
50        bins B_data_0       = {0};
51        bins B_data_max     = {MAXPOS};
52        bins B_data_min     = {MAXNEG};
53        bins B_data_default = default;
54      }
55
56      // Opcode Coverpoint
57      coverpoint opcode {
58        bins bins_shift[]   = {SHIFT_OP, ROT_OP};
59        bins bins_arith[]   = {ADD_OP, MUL_OP};
60        bins bins_bitwise[] = {OR_OP, XOR_OP};
61        illegal_bins bins_invalid = {INVALID_6, INVALID_7};
62      }
63    endgroup
64
```

```
65      // -------- Constraints --------
66      constraint c1 { if (en_c1) A inside {[MAXNEG:MAXPOS]}; }
67      constraint c2 { if (en_c2) B inside {[MAXNEG:MAXPOS]}; }
68      constraint c3 { if (en_c3) opcode inside {OR_OP, XOR_OP, ADD_OP, MUL_OP, SHIFT_OP, ROT_OP}; }
69
70      // Constraint #8: unique opcodes in array
71      constraint unique_opcodes {
72        if (en_c8) {
73          foreach (opcode_seq[i]) opcode_seq[i] inside {OR_OP, XOR_OP, ADD_OP, MUL_OP, SHIFT_OP, ROT_OP};
74          unique {opcode_seq};
75        }
76      }
77
78      // -------- Constructor --------
79      function new();
80        en_c1 = 1; en_c2 = 1; en_c3 = 1; en_c4 = 1;
81        en_c5 = 1; en_c6 = 1; en_c7 = 1; en_c8 = 0;
82        cvr_gp = new();
83      endfunction
84
85    endclass : alsu_txn
86
87  endpackage : alsu_pkg
88
```

3. Design code

```
module ALSU(A, B, cin, serial_in, red_op_A, red_op_B, opcode, bypass_A, bypass_B, clk, rst, direction, leds, out);
parameter INPUT_PRIORITY = "A";
parameter FULL_ADDER = "ON";
input clk, cin, rst, red_op_A, red_op_B, bypass_A, bypass_B, direction, serial_in;
input [2:0] opcode;
input signed [2:0] A, B;
output reg [15:0] leds;
output reg signed [5:0] out;

reg red_op_A_reg, red_op_B_reg, bypass_A_reg, bypass_B_reg, direction_reg, serial_in_reg;
reg signed [1:0] cin_reg;
reg [2:0] opcode_reg;
reg signed [2:0] A_reg, B_reg;

wire invalid_red_op, invalid_opcode, invalid;

//Invalid handling
assign invalid_red_op = (red_op_A_reg | red_op_B_reg) & (opcode_reg[1] | opcode_reg[2]);
assign invalid_opcode = opcode_reg[1] & opcode_reg[2];
assign invalid = invalid_red_op | invalid_opcode;

//Registering input signals
always @(posedge clk or posedge rst) begin
  if(rst) begin
      cin_reg <= 0;
      red_op_B_reg <= 0;
      red_op_A_reg <= 0;
      bypass_B_reg <= 0;
      bypass_A_reg <= 0;
      direction_reg <= 0;
      serial_in_reg <= 0;
      opcode_reg <= 0;
      A_reg <= 0;
      B_reg <= 0;
  end else begin
      cin_reg <= cin;
```

```
37          red_op_B_reg <= red_op_B;
38          red_op_A_reg <= red_op_A;
39          bypass_B_reg <= bypass_B;
40          bypass_A_reg <= bypass_A;
41          direction_reg <= direction;
42          serial_in_reg <= serial_in;
43          opcode_reg <= opcode;
44          A_reg <= A;
45          B_reg <= B;
46      end
47  end
48
49  //leds output blinking
50  always @(posedge clk or posedge rst) begin
51      if(rst) begin
52          leds <= 0;
53      end else begin
54          if (invalid)
55              leds <= ~leds;
56          else
57              leds <= 0;
58      end
59  end
60
61  //ALSU output processing
62  always @(posedge clk or posedge rst) begin
63      if(rst) begin
64          out <= 0;
65      end
66      else begin
67          if (bypass_A_reg && bypass_B_reg)
68              out <= (INPUT_PRIORITY == "A")? A_reg: B_reg;
69          else if (bypass_A_reg)
```

```
69      else if (bypass_A_reg)
70        out <= A_reg;
71      else if (bypass_B_reg)
72        out <= B_reg;
73      else if (invalid)
74          out <= 0;
75      else begin
76          case (opcode_reg)  //edited design error
77            3'h0: begin
78              if (red_op_A_reg && red_op_B_reg)
79                out <= (INPUT_PRIORITY == "A")? |A_reg: |B_reg;
80              else if (red_op_A_reg)
81                out <= |A_reg;
82              else if (red_op_B_reg)
83                out <= |B_reg;
84              else
85                out <= A_reg | B_reg;
86            end
87            3'h1: begin
88              if (red_op_A_reg && red_op_B_reg)
89                out <= (INPUT_PRIORITY == "A")? ^A_reg: ^B_reg;
90              else if (red_op_A_reg)
91                out <= ^A_reg;
92              else if (red_op_B_reg)
93                out <= ^B_reg;
94              else
95                out <= A_reg ^ B_reg;
96            end
97            3'h2: out <= (FULL_ADDER == "ON") ? (A_reg + B_reg + cin_reg) : (A_reg + B_reg);  //edited design error
98            3'h3: out <= A_reg * B_reg;
99            3'h4: begin
100             if (direction_reg)
101               out <= {out[4:0], serial_in_reg};
```

```
100             if (direction_reg)
101               out <= {out[4:0], serial_in_reg};
102             else
103               out <= {serial_in_reg, out[5:1]};
104           end
105           3'h5: begin
106             if (direction_reg)
107               out <= {out[4:0], out[5]};
108             else
109               out <= {out[0], out[5:1]};
110           end
111           default: out <= 0; //edit for 100% coverage and exclude it
112
113         endcase
114       end
115     end
116   end
117
118   endmodule
```

4. Report any bugs detected in the design and fix them

5. Snippet to your verification plan document

Label	Design Requirement Description	Stimulus Generation	Functional Coverage	Functionality Check
ALSU_1	When rst is asserted, both out and leds should be reset to 0.	Directed at the start of the simulation.	-	A checker in the testbench ensures out == 0 and leds == 0 when rst == 1.
ALSU_2	When bypass_A or bypass_B are asserted, out should directly reflect the corresponding input (A or B).	Randomization of bypass_A and bypass_B.	Cover bins for cases where bypass_A == 1, bypass_B == 1, and both asserted.	Checker ensures out == A or out == B according to priority when bypass is active.
ALSU_3	If both bypass_A and bypass_B are asserted, priority is given to A when INPUT_PRIORITY parameter is "A".	Directed stimulus where both bypass signals are high.	Cover bins for both bypass signals high and INPUT_PRIORITY set to "A".	Checker ensures out == A when both bypass signals are asserted.
ALSU_4	When opcode corresponds to bitwise OR (3'h0), the output must equal A OR B.	Randomized values for A, B, and opcode.	Cover bins for opcode 3'h0.	Checker compares DUT output to A OR B.
ALSU_5	When opcode corresponds to bitwise XOR (3'h1), the output must equal A XOR B.	Randomized values for A, B, and opcode.	Cover bins for opcode 3'h1.	Checker compares DUT output to A XOR B.
ALSU_6	When opcode corresponds to ADD (3'h2), output should be A + B + cin if FULL_ADDER parameter is "ON".	Randomized A, B, and cin with opcode == 3'h2.	Cover bins for ADD operation with different cin values.	Checker compares DUT output to expected A + B + cin.
ALSU_7	When opcode corresponds to MUL (3'h3), output should be A * B.	Randomized A and B with opcode == 3'h3.	Cover bins for MUL operation.	Checker compares DUT output to expected A * B.
ALSU_8	When opcode corresponds to SHIFT (3'h4), out should shift left or right depending on direction and serial_in.	Directed and randomized tests toggling direction and serial_in.	Cover bins for shift left and shift right operations.	Checker verifies shift behavior bit-by-bit.

ALSU_9	When opcode corresponds to ROTATE (3'h5), out should rotate left or right based on direction.	Directed and randomized tests toggling direction.	Cover bins for rotate left and rotate right operations.	Checker verifies rotation behavior bit-by-bit.
ALSU_10	When opcode is invalid (3'h6 or 3'h7), out should be 0 and LEDs should blink.	Directed test using invalid opcodes.	Cover bins for invalid opcode values.	Checker ensures out == 0 and leds toggles when invalid opcode is detected.
ALSU_11	When both red_op_A and red_op_B are asserted, priority is given to A when INPUT_PRIORITY is "A".	Randomized with both reduction signals high.	Cover bins for both reduction signals high.	Checker ensures output matches reduction of A only.
ALSU_12	Verify walking ones patterns for A and B according to red_op_A and red_op_B rules.	Randomized inputs with special handling to generate walking ones 001, 010, 100.	Cover bins for A_data_walkingones [] and B_data_walkingones [].	Checker verifies outputs match reduction rules when walking ones are applied.
ALSU_13	Validate transition of opcodes in the sequence 0 -> 1 -> 2 -> 3 -> 4 -> 5.	Constraint 8 to generate a unique fixed array of opcodes.	Cover transition bins for the specified opcode sequence.	Checker verifies that opcodes occur in the specified sequence and behavior matches expectations.
ALSU_14	Verify LED blinking behavior whenever invalid signal is high.	Directed test toggling invalid condition.	Cover bins for LED state toggling.	Checker verifies LEDs toggle at every clock edge while invalid is high.

6. Do file

```
vlib work
vlog ALSU.v package_q_3.svh question_3_tb.svh  +cover -covercells
vsim -voptargs=+acc work.question_3_tb -cover
add wave *
coverage save question_3_tb.ucdb -onexit
run -all
```

7. Code Coverage report snippets (Questions 1, 2, and 3 only)

code_cvr_alu - Notepad

File Edit Format View Help

```
Coverage Report by instance with details

================================================================================
=== Instance: /question_3_tb/DUT
=== Design Unit: work.ALSU
================================================================================
Branch Coverage:
    Enabled Coverage              Bins      Hits    Misses  Coverage
    ----------------              ----      ----    ------  --------
    Branches                        31        31         0   100.00%

================================Branch Details================================

Branch Coverage for instance /question_3_tb/DUT

    Line         Item                      Count     Source
    ----         ----                      -----     ------
  File ALSU.v
------------------------------------IF Branch------------------------------------
    24                                     1038     Count coming in to IF
    24              1                         2       if(rst) begin

    35              1                      1036       end else begin

Branch totals: 2 hits of 2 branches = 100.00%

------------------------------------IF Branch------------------------------------
    51                                     1039     Count coming in to IF
    51              1                         3       if(rst) begin

    53              1                      1036       end else begin

Branch totals: 2 hits of 2 branches = 100.00%

------------------------------------IF Branch------------------------------------
    54                                     1036     Count coming in to IF
    54              1                       511           if (invalid)

    56              1                       525           else

Branch totals: 2 hits of 2 branches = 100.00%
```

code_cvr_alu - Notepad

File Edit Format View Help

```
-----------------------------------IF Branch-----------------------------------
    63                                  1038     Count coming in to IF
    63              1                      2       if(rst) begin

    66              1                   1036       else begin

Branch totals: 2 hits of 2 branches = 100.00%

-----------------------------------IF Branch-----------------------------------
    67                                  1036     Count coming in to IF
    67              1                    226         if (bypass_A_reg && bypass_B_reg)

    69              1                    262         else if (bypass_A_reg)

    71              1                    249         else if (bypass_B_reg)

    73              1                    134         else if (invalid)

    75              1                    165         else begin

Branch totals: 5 hits of 5 branches = 100.00%

-----------------------------------CASE Branch-----------------------------------
    76                                  165     Count coming in to CASE
    77              1                    50              3'h0: begin

    87              1                    44              3'h1: begin

    97              1                    15              3'h2: out <= (FULL_ADDER == "ON") ? (A_reg + B_reg + cin_reg) : (A_reg + B_reg);  //edited design error

    98              1                    15              3'h3: out <= A_reg * B_reg;

    99              1                    27              3'h4: begin

    105             1                    14              3'h5: begin

Branch totals: 6 hits of 6 branches = 100.00%

-----------------------------------IF Branch-----------------------------------
    78                                   50     Count coming in to IF
    78              1                     7               if (red_op_A_reg && red_op_B_reg)

    80              1                     8               else if (red_op_A_reg)
```

code_cvr_alu - Notepad

File Edit Format View Help

```
    84              1                          26                else

Branch totals: 4 hits of 4 branches = 100.00%

------------------------------------IF Branch------------------------------------
    88                                         44     Count coming in to IF
    88              1                          10                if (red_op_A_reg && red_op_B_reg)

    90              1                          13                else if (red_op_A_reg)

    92              1                           9                else if (red_op_B_reg)

    94              1                          12                else

Branch totals: 4 hits of 4 branches = 100.00%

------------------------------------IF Branch------------------------------------
    100                                        27     Count coming in to IF
    100             1                          13                if (direction_reg)

    102             1                          14                else

Branch totals: 2 hits of 2 branches = 100.00%

------------------------------------IF Branch------------------------------------
    106                                        14     Count coming in to IF
    106             1                          10                if (direction_reg)

    108             1                           4                else

Branch totals: 2 hits of 2 branches = 100.00%


Condition Coverage:
    Enabled Coverage              Bins   Covered    Misses  Coverage
    ----------------              ----      ----    ------  --------
    Conditions                       6         6         0   100.00%

================================Condition Details================================

Condition Coverage for instance /question_3_tb/DUT --
```

code_cvr_alu - Notepad

File Edit Format View Help

```
================================Condition Details================================

Condition Coverage for instance /question_3_tb/DUT --

  File ALSU.v
----------------Focused Condition View-------------------
Line       67 Item    1  (bypass_A_reg && bypass_B_reg)
Condition totals: 2 of 2 input terms covered = 100.00%

    Input Term   Covered  Reason for no coverage   Hint
   -----------  --------  ----------------------  --------------
  bypass_A_reg         Y
  bypass_B_reg         Y

     Rows:       Hits  FEC Target            Non-masking condition(s)
 ---------  ---------  --------------------  -------------------------
  Row   1:          1  bypass_A_reg_0        -
  Row   2:          1  bypass_A_reg_1        bypass_B_reg
  Row   3:          1  bypass_B_reg_0        bypass_A_reg
  Row   4:          1  bypass_B_reg_1        bypass_A_reg

----------------Focused Condition View-------------------
Line       78 Item    1  (red_op_A_reg && red_op_B_reg)
Condition totals: 2 of 2 input terms covered = 100.00%

    Input Term   Covered  Reason for no coverage   Hint
   -----------  --------  ----------------------  --------------
  red_op_A_reg         Y
  red_op_B_reg         Y

     Rows:       Hits  FEC Target            Non-masking condition(s)
 ---------  ---------  --------------------  -------------------------
  Row   1:          1  red_op_A_reg_0        -
  Row   2:          1  red_op_A_reg_1        red_op_B_reg
  Row   3:          1  red_op_B_reg_0        red_op_A_reg
  Row   4:          1  red_op_B_reg_1        red_op_A_reg

----------------Focused Condition View-------------------
Line       88 Item    1  (red_op_A_reg && red_op_B_reg)
Condition totals: 2 of 2 input terms covered = 100.00%

    Input Term   Covered  Reason for no coverage   Hint
   -----------  --------  ----------------------  --------------
```

code_cvr_alu - Notepad

File Edit Format View Help

```
Condition totals: 2 of 2 input terms covered = 100.00%

    Input Term   Covered  Reason for no coverage   Hint
   -----------  --------  ----------------------  --------------
  red_op_A_reg         Y
  red_op_B_reg         Y

     Rows:       Hits  FEC Target            Non-masking condition(s)
 ---------  ---------  --------------------  -------------------------
  Row   1:          1  red_op_A_reg_0        -
  Row   2:          1  red_op_A_reg_1        red_op_B_reg
  Row   3:          1  red_op_B_reg_0        red_op_A_reg
  Row   4:          1  red_op_B_reg_1        red_op_A_reg


Expression Coverage:
    Enabled Coverage              Bins   Covered    Misses  Coverage
    ----------------              ----      ----    ------  --------
    Expressions                      8         5         3    62.50%

================================Expression Details================================

Expression Coverage for instance /question_3_tb/DUT --

  File ALSU.v
----------------Focused Expression View-----------------
Line       18 Item    1  ((red_op_A_reg | red_op_B_reg) & (opcode_reg[1] | opcode_reg[2]))
Expression totals: 4 of 4 input terms covered = 100.00%

     Input Term   Covered  Reason for no coverage   Hint
    -----------  --------  ----------------------  --------------
   red_op_A_reg         Y
   red_op_B_reg         Y
  opcode_reg[1]         Y
  opcode_reg[2]         Y

     Rows:       Hits  FEC Target            Non-masking condition(s)
 ---------  ---------  --------------------  -------------------------
  Row   1:          1  red_op_A_reg_0        ((opcode_reg[1] | opcode_reg[2]) && ~red_op_B_reg)
  Row   2:          1  red_op_A_reg_1        ((opcode_reg[1] | opcode_reg[2]) && ~red_op_B_reg)
  Row   3:          1  red_op_B_reg_0        ((opcode_reg[1] | opcode_reg[2]) && ~red_op_A_reg)
  Row   4:          1  red_op_B_reg_1        ((opcode_reg[1] | opcode_reg[2]) && ~red_op_A_reg)
  Row   5:          1  opcode_reg[1]_0       ((red_op_A_reg | red_op_B_reg) && ~opcode_reg[2])
```

code_cvr_alu - Notepad

File Edit Format View Help

```
  Row   5:          1  opcode_reg[1]_0       ((red_op_A_reg | red_op_B_reg) && ~opcode_reg[2])
  Row   6:          1  opcode_reg[1]_1       ((red_op_A_reg | red_op_B_reg) && ~opcode_reg[2])
  Row   7:          1  opcode_reg[2]_0       ((red_op_A_reg | red_op_B_reg) && ~opcode_reg[1])
  Row   8:          1  opcode_reg[2]_1       ((red_op_A_reg | red_op_B_reg) && ~opcode_reg[1])

----------------Focused Expression View-----------------
Line       19 Item    1  (opcode_reg[1] & opcode_reg[2])
Expression totals: 0 of 2 input terms covered = 0.00%

     Input Term   Covered  Reason for no coverage   Hint
    -----------  --------  -----------------------  --------------
  opcode_reg[1]         N  '_1' not hit             Hit '_1'
  opcode_reg[2]         N  '_1' not hit             Hit '_1'

     Rows:       Hits  FEC Target            Non-masking condition(s)
 ---------  ---------  --------------------  -------------------------
  Row   1:          1  opcode_reg[1]_0       opcode_reg[2]
  Row   2:    ***0***  opcode_reg[1]_1       opcode_reg[2]
  Row   3:          1  opcode_reg[2]_0       opcode_reg[1]
  Row   4:    ***0***  opcode_reg[2]_1       opcode_reg[1]

----------------Focused Expression View-----------------
Line       20 Item    1  (invalid_red_op | invalid_opcode)
Expression totals: 1 of 2 input terms covered = 50.00%

      Input Term   Covered  Reason for no coverage   Hint
     -----------  --------  -----------------------  --------------
  invalid_red_op         Y
  invalid_opcode         N  '_1' not hit             Hit '_1'

     Rows:       Hits  FEC Target            Non-masking condition(s)
 ---------  ---------  --------------------  -------------------------
  Row   1:          1  invalid_red_op_0      ~invalid_opcode
  Row   2:          1  invalid_red_op_1      ~invalid_opcode
  Row   3:          1  invalid_opcode_0      ~invalid_red_op
  Row   4:    ***0***  invalid_opcode_1      ~invalid_red_op


Statement Coverage:
    Enabled Coverage              Bins      Hits    Misses  Coverage
    ----------------              ----      ----    ------  --------
    Statements                      48        48         0   100.00%
```

code_cvr_alu - Notepad

File Edit Format View Help

```
================================Statement Details================================

Statement Coverage for instance /question_3_tb/DUT --

    Line         Item                      Count     Source
    ----         ----                      -----     ------
  File ALSU.v
    1                                                module ALSU(A, B, cin, serial_in, red_op_A, red_op_B, opcode, bypass_A, bypass_B, clk, rst, direction, leds, out);
    2                                                parameter INPUT_PRIORITY = "A";
    3                                                parameter FULL_ADDER = "ON";
    4                                                input clk, cin, rst, red_op_A, red_op_B, bypass_A, bypass_B, direction, serial_in;
    5                                                input [2:0] opcode;
    6                                                input signed [2:0] A, B;
    7                                                output reg [15:0] leds;
    8                                                output reg signed [5:0] out;
    9
    10                                               reg red_op_A_reg, red_op_B_reg, bypass_A_reg, bypass_B_reg, direction_reg, serial_in_reg;
    11                                               reg signed [1:0] cin_reg;
    12                                               reg [2:0] opcode_reg;
    13                                               reg signed [2:0] A_reg, B_reg;
    14
    15                                               wire invalid_red_op, invalid_opcode, invalid;
    16
    17                                               //Invalid handling
    18              1                         973    assign invalid_red_op = (red_op_A_reg | red_op_B_reg) & (opcode_reg[1] | opcode_reg[2]);
```

code_cvr_alu - Notepad

File Edit Format View Help

```
    18              1                         973    assign invalid_red_op = (red_op_A_reg | red_op_B_reg) & (opcode_reg[1] | opcode_reg[2]);
    19              1                         840    assign invalid_opcode = opcode_reg[1] & opcode_reg[2];
    20              1                         526    assign invalid = invalid_red_op | invalid_opcode;
    21
    22                                               //Registering input signals
    23              1                        1038    always @(posedge clk or posedge rst) begin
    24                                                 if(rst) begin
    25              1                           2        cin_reg <= 0;
    26              1                           2        red_op_B_reg <= 0;
    27              1                           2        red_op_A_reg <= 0;
    28              1                           2        bypass_B_reg <= 0;
    29              1                           2        bypass_A_reg <= 0;
    30              1                           2        direction_reg <= 0;
    31              1                           2        serial_in_reg <= 0;
    32              1                           2        opcode_reg <= 0;
    33              1                           2        A_reg <= 0;
    34              1                           2        B_reg <= 0;
    35                                                 end else begin
    36              1                        1036        cin_reg <= cin;
    37              1                        1036        red_op_B_reg <= red_op_B;
    38              1                        1036        red_op_A_reg <= red_op_A;
    39              1                        1036        bypass_B_reg <= bypass_B;
```

code_cvr_alu - Notepad

File Edit Format View Help

39	1	1036	bypass_B_reg <= bypass_B;
40	1	1036	bypass_A_reg <= bypass_A;
41	1	1036	direction_reg <= direction;
42	1	1036	serial_in_reg <= serial_in;
43	1	1036	opcode_reg <= opcode;
44	1	1036	A_reg <= A;
45	1	1036	B_reg <= B;
46			end
47			end
48			
49			//leds output blinking
50	1	1039	always @(posedge clk or posedge rst) begin
51			if(rst) begin
52	1	3	leds <= 0;
53			end else begin
54			if (invalid)
55	1	511	leds <= ~leds;
56			else
57	1	525	leds <= 0;
58			end
59			end
60			

code_cvr_alu - Notepad

File Edit Format View Help

```
60
61                                          //ALSU output processing
62              1                  1038     always @(posedge clk or posedge rst) begin
63                                            if(rst) begin
64              1                     2        out <= 0;
65                                            end
66                                            else begin
67                                              if (bypass_A_reg && bypass_B_reg)
68              1                   226          out <= (INPUT_PRIORITY == "A")? A_reg: B_reg;
69                                              else if (bypass_A_reg)
70              1                   262          out <= A_reg;
71                                              else if (bypass_B_reg)
72              1                   249          out <= B_reg;
73                                              else if (invalid)
74              1                   134            out <= 0;
75                                              else begin
76                                                  case (opcode_reg)  //edited design error
77                                                    3'h0: begin
78                                                      if (red_op_A_reg && red_op_B_reg)
79              1                     7                   out <= (INPUT_PRIORITY == "A")? |A_reg: |B_reg;
80                                                      else if (red_op_A_reg)
81              1                     8                   out <= |A_reg;
```

code_cvr_alu - Notepad

File Edit Format View Help

```
81              1                       8                 out <= |A_reg;
82                                                      else if (red_op_B_reg)
83              1                       9                 out <= |B_reg;
84                                                      else
85              1                      26                 out <= A_reg | B_reg;
86                                                    end
87                                                    3'h1: begin
88                                                      if (red_op_A_reg && red_op_B_reg)
89              1                      10                 out <= (INPUT_PRIORITY == "A")? ^A_reg: ^B_reg;
90                                                      else if (red_op_A_reg)
91              1                      13                 out <= ^A_reg;
92                                                      else if (red_op_B_reg)
93              1                       9                 out <= ^B_reg;
94                                                      else
95              1                      12                 out <= A_reg ^ B_reg;
96                                                    end
97              1                      15                 3'h2: out <= (FULL_ADDER == "ON") ? (A_reg + B_reg + cin_reg) : (A_reg + B_reg);  //edited design error
98              1                      15                 3'h3: out <= A_reg * B_reg;
99                                                    3'h4: begin
100                                                     if (direction_reg)
101             1                      13                 out <= {out[4:0], serial_in_reg};
102                                                     else
```

code_cvr_alu - Notepad

File Edit Format View Help

```
102                                                     else
103             1                      14                 out <= {serial_in_reg, out[5:1]};
104                                                   end
105                                                   3'h5: begin
106                                                     if (direction_reg)
107             1                      10                 out <= {out[4:0], out[5]};
108                                                     else
109             1                       4                 out <= {out[0], out[5:1]};
```

Toggle Coverage:

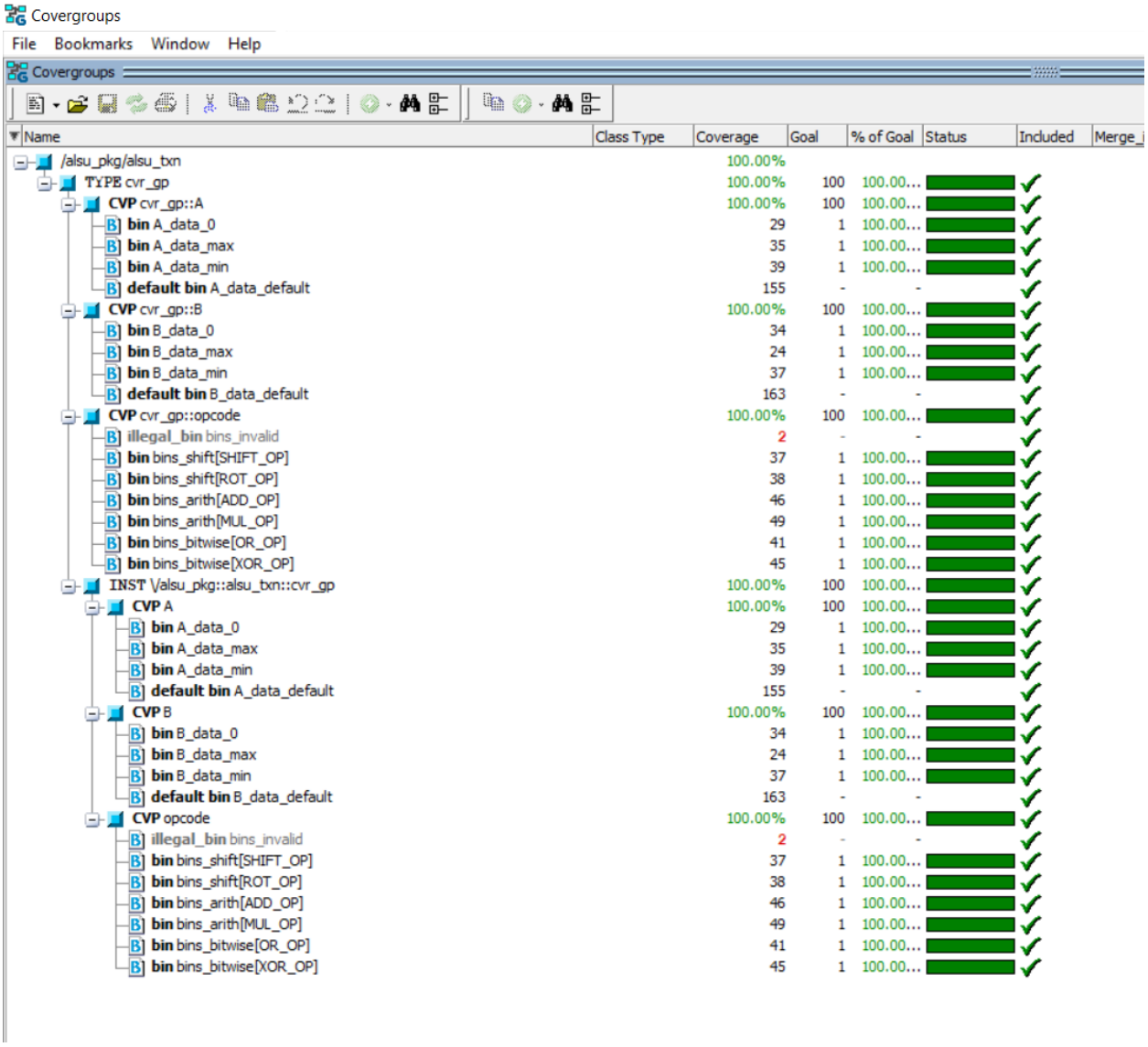
Enabled Coverage	Bins	Hits	Misses	Coverage
Toggles	114	114	0	100.00%

code_cvr_alu - Notepad

File Edit Format View Help

```
    Enabled Coverage              Bins      Hits    Misses  Coverage
    ----------------              ----      ----    ------  --------
    Toggles                        114       114         0   100.00%

================================Toggle Details================================

Toggle Coverage for instance /question_3_tb/DUT --

                                          Node      1H->0L      0L->1H                          "Coverage"
                                          ----------------------------------------------------------------
                                         A[0-2]           1           1                             100.00
                                     A_reg[2-0]           1           1                             100.00
                                         B[0-2]           1           1                             100.00
                                     B_reg[2-0]           1           1                             100.00
                                       bypass_A           1           1                             100.00
                                   bypass_A_reg           1           1                             100.00
                                       bypass_B           1           1                             100.00
                                   bypass_B_reg           1           1                             100.00
                                            cin           1           1                             100.00
                                      cin_reg[0]           1           1                             100.00
                                            clk           1           1                             100.00
                                      direction           1           1                             100.00
                                  direction_reg           1           1                             100.00
                                        invalid           1           1                             100.00
                                 invalid_red_op           1           1                             100.00
                                     leds[15-0]           1           1                             100.00
                                    opcode[0-2]           1           1                             100.00
                                opcode_reg[2-0]           1           1                             100.00
                                       out[5-0]           1           1                             100.00
                                       red_op_A           1           1                             100.00
                                   red_op_A_reg           1           1                             100.00
                                       red_op_B           1           1                             100.00
                                   red_op_B_reg           1           1                             100.00
                                      serial_in           1           1                             100.00
                                  serial_in_reg           1           1                             100.00

Total Node Count     =         57
Toggled Node Count   =         57
Untoggled Node Count =          0

Toggle Coverage      =     100.00% (114 of 114 bins)

================================================================================
```

8. Functional Coverage report snippets (Questions 1, 2, and 3 only)


Covergroups
File Bookmarks Window Help
Covergroups
Name
Class Type
Coverage
Goal
% of Goal
Status
Included
Merge_i
/alsu_pkg/alsu_txn 100.00%
TYPE cvr_gp 100.00% 100 100.00...
CVP cvr_gp::A 100.00% 100 100.00...
bin A_data_0 29 1 100.00...
bin A_data_max 35 1 100.00...
bin A_data_min 39 1 100.00...
default bin A_data_default 155 - -
CVP cvr_gp::B 100.00% 100 100.00...
bin B_data_0 34 1 100.00...
bin B_data_max 24 1 100.00...
bin B_data_min 37 1 100.00...
default bin B_data_default 163 - -
CVP cvr_gp::opcode 100.00% 100 100.00...
illegal_bin bins_invalid 2 - -
bin bins_shift[SHIFT_OP] 37 1 100.00...
bin bins_shift[ROT_OP] 38 1 100.00...
bin bins_arith[ADD_OP] 46 1 100.00...
bin bins_arith[MUL_OP] 49 1 100.00...
bin bins_bitwise[OR_OP] 41 1 100.00...
bin bins_bitwise[XOR_OP] 45 1 100.00...
INST \/alsu_pkg::alsu_txn::cvr_gp 100.00% 100 100.00...
CVP A 100.00% 100 100.00...
bin A_data_0 29 1 100.00...
bin A_data_max 35 1 100.00...
bin A_data_min 39 1 100.00...
default bin A_data_default 155 - -
CVP B 100.00% 100 100.00...
bin B_data_0 34 1 100.00...
bin B_data_max 24 1 100.00...
bin B_data_min 37 1 100.00...
default bin B_data_default 163 - -
CVP opcode 100.00% 100 100.00...
illegal_bin bins_invalid 2 - -
bin bins_shift[SHIFT_OP] 37 1 100.00...
bin bins_shift[ROT_OP] 38 1 100.00...
bin bins_arith[ADD_OP] 46 1 100.00...
bin bins_arith[MUL_OP] 49 1 100.00...
bin bins_bitwise[OR_OP] 41 1 100.00...
bin bins_bitwise[XOR_OP] 45 1 100.00...

code_cvr_alu - Notepad

File Edit Format View Help

```
Covergroup Coverage:
    Covergroups                     1        na        na   100.00%
        Coverpoints/Crosses         3        na        na        na
            Covergroup Bins        12        12         0   100.00%
------------------------------------------------------------------------------------------------------------------------
Covergroup                                          Metric       Goal       Bins    Status

------------------------------------------------------------------------------------------------------------------------
 TYPE /alsu_pkg/alsu_txn/cvr_gp                   100.00%        100          -    Covered
    covered/total bins:                                12         12          -
    missing/total bins:                                 0         12          -
    % Hit:                                         100.00%        100          -
    Coverpoint A                                   100.00%        100          -    Covered
        covered/total bins:                             3          3          -
        missing/total bins:                             0          3          -
        % Hit:                                     100.00%        100          -
        bin A_data_0                                   29          1          -    Covered
        bin A_data_max                                 35          1          -    Covered
        bin A_data_min                                 39          1          -    Covered
        default bin A_data_default                    155                     -    Occurred
    Coverpoint B                                   100.00%        100          -    Covered
        covered/total bins:                             3          3          -
        missing/total bins:                             0          3          -
        % Hit:                                     100.00%        100          -
        bin B_data_0                                   34          1          -    Covered
        bin B_data_max                                 24          1          -    Covered
        bin B_data_min                                 37          1          -    Covered
        default bin B_data_default                    163                     -    Occurred
    Coverpoint opcode                              100.00%        100          -    Covered
        covered/total bins:                             6          6          -
        missing/total bins:                             0          6          -
        % Hit:                                     100.00%        100          -
        illegal_bin bins_invalid                        2                     -    Occurred
        bin bins_shift[SHIFT_OP]                       37          1          -    Covered
        bin bins_shift[ROT_OP]                         38          1          -    Covered
        bin bins_arith[ADD_OP]                         46          1          -    Covered
        bin bins_arith[MUL_OP]                         49          1          -    Covered
        bin bins_bitwise[OR_OP]                        41          1          -    Covered
        bin bins_bitwise[XOR_OP]                       45          1          -    Covered
 Covergroup instance \/alsu_pkg::alsu_txn::cvr_gp 100.00%        100          -    Covered
    covered/total bins:                                12         12          -
    missing/total bins:                                 0         12          -
    % Hit:                                         100.00%        100          -
```

code_cvr_alu - Notepad

File Edit Format View Help

```
    % Hit:                                  100.00%        100          -
    Coverpoint A                            100.00%        100          -    Covered
        covered/total bins:                       3          3          -
        missing/total bins:                       0          3          -
        % Hit:                              100.00%        100          -
        bin A_data_0                             29          1          -    Covered
        bin A_data_max                           35          1          -    Covered
        bin A_data_min                           39          1          -    Covered
        default bin A_data_default              155                     -    Occurred
    Coverpoint B                            100.00%        100          -    Covered
        covered/total bins:                       3          3          -
        missing/total bins:                       0          3          -
        % Hit:                              100.00%        100          -
        bin B_data_0                             34          1          -    Covered
        bin B_data_max                           24          1          -    Covered
        bin B_data_min                           37          1          -    Covered
        default bin B_data_default              163                     -    Occurred
    Coverpoint opcode                       100.00%        100          -    Covered
        covered/total bins:                       6          6          -
        missing/total bins:                       0          6          -
        % Hit:                              100.00%        100          -
        illegal_bin bins_invalid                  2                     -    Occurred
        bin bins_shift[SHIFT_OP]                 37          1          -    Covered
        bin bins_shift[ROT_OP]                   38          1          -    Covered
        bin bins_arith[ADD_OP]                   46          1          -    Covered
        bin bins_arith[MUL_OP]                   49          1          -    Covered
        bin bins_bitwise[OR_OP]                  41          1          -    Covered
        bin bins_bitwise[XOR_OP]                 45          1          -    Covered
Statement Coverage:
    Enabled Coverage              Bins      Hits    Misses  Coverage
    ----------------              ----      ----    ------  --------
    Statements                       9         9         0   100.00%
```

code_cvr_alu - Notepad

File Edit Format View Help

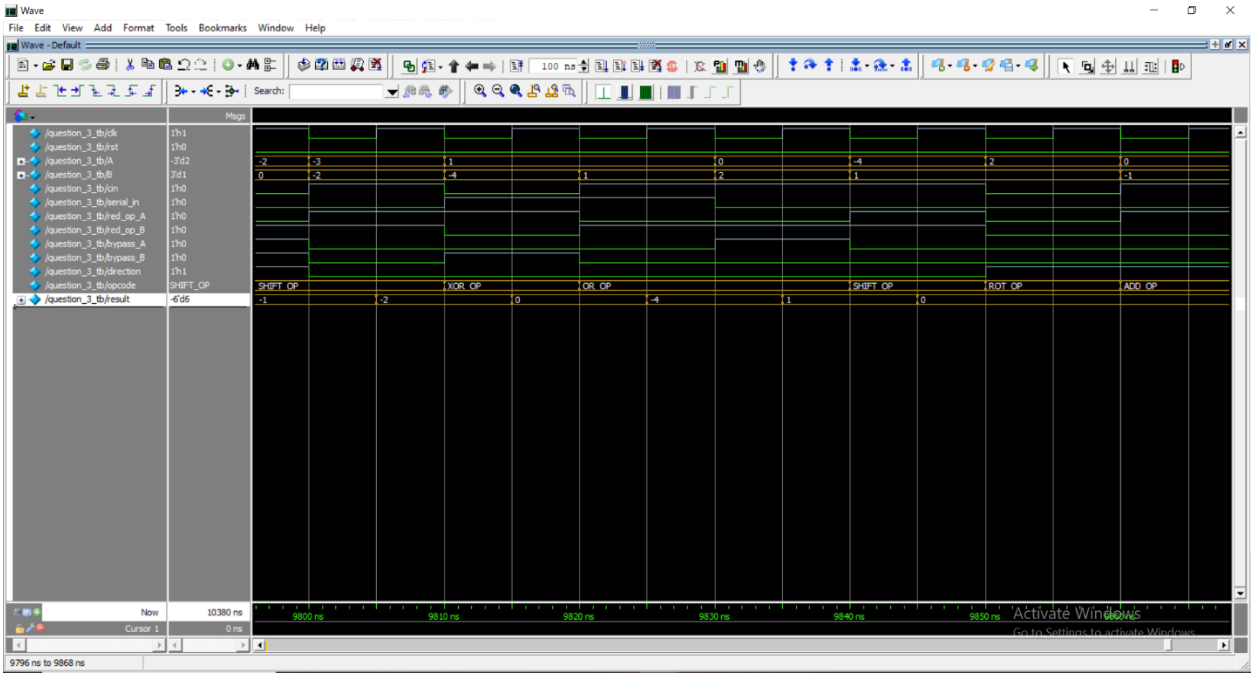
Covergroup	Metric	Goal	Bins	Status
TYPE /alsu_pkg/alsu_txn/cvr_gp	100.00%	100	-	Covered
covered/total bins:	12	12	-	
missing/total bins:	0	12	-	
% Hit:	100.00%	100	-	
Coverpoint A	100.00%	100	-	Covered
covered/total bins:	3	3	-	
missing/total bins:	0	3	-	
% Hit:	100.00%	100	-	
bin A_data_0	29	1	-	Covered
bin A_data_max	35	1	-	Covered
bin A_data_min	39	1	-	Covered
default bin A_data_default	155		-	Occurred
Coverpoint B	100.00%	100	-	Covered
covered/total bins:	3	3	-	
missing/total bins:	0	3	-	
% Hit:	100.00%	100	-	
bin B_data_0	34	1	-	Covered
bin B_data_max	24	1	-	Covered
bin B_data_min	37	1	-	Covered
default bin B_data_default	163		-	Occurred
Coverpoint opcode	100.00%	100	-	Covered
covered/total bins:	6	6	-	
missing/total bins:	0	6	-	
% Hit:	100.00%	100	-	
illegal_bin bins_invalid	2		-	Occurred
bin bins_shift[SHIFT_OP]	37	1	-	Covered
bin bins_shift[ROT_OP]	38	1	-	Covered
bin bins_arith[ADD_OP]	46	1	-	Covered
bin bins_arith[MUL_OP]	49	1	-	Covered
bin bins_bitwise[OR_OP]	41	1	-	Covered
bin bins_bitwise[XOR_OP]	45	1	-	Covered
Covergroup instance \/alsu_pkg::alsu_txn::cvr_gp	100.00%	100	-	Covered
covered/total bins:	12	12	-	
missing/total bins:	0	12	-	
% Hit:	100.00%	100	-	
Coverpoint A	100.00%	100	-	Covered
covered/total bins:	3	3	-	
missing/total bins:	0	3	-	
% Hit:	100.00%	100	-	

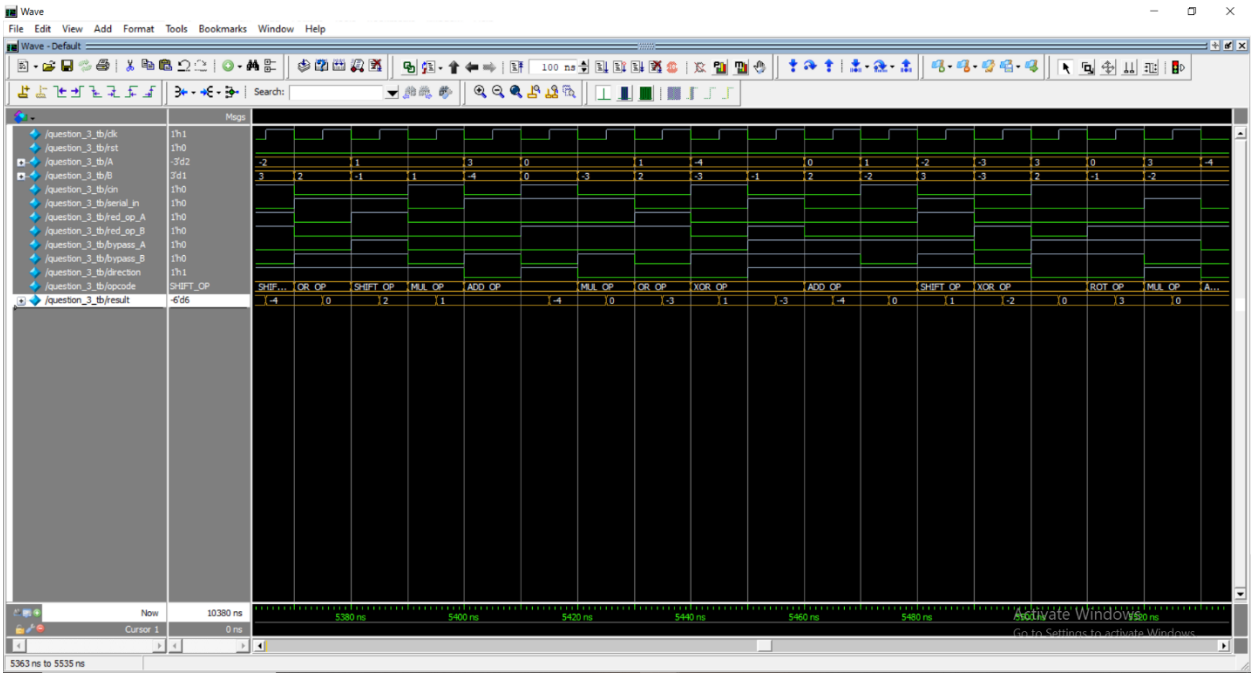
code_cvr_alu - Notepad

File Edit Format View Help

```
        bin bins_shift[SHIFT_OP]                          37          1          -    Covered
        bin bins_shift[ROT_OP]                            38          1          -    Covered
        bin bins_arith[ADD_OP]                            46          1          -    Covered
        bin bins_arith[MUL_OP]                            49          1          -    Covered
        bin bins_bitwise[OR_OP]                           41          1          -    Covered
        bin bins_bitwise[XOR_OP]                          45          1          -    Covered
 Covergroup instance \/alsu_pkg::alsu_txn::cvr_gp     100.00%        100          -    Covered
    covered/total bins:                                    12         12          -
    missing/total bins:                                     0         12          -
    % Hit:                                             100.00%        100          -
    Coverpoint A                                       100.00%        100          -    Covered
        covered/total bins:                                 3          3          -
        missing/total bins:                                 0          3          -
        % Hit:                                         100.00%        100          -
        bin A_data_0                                       29          1          -    Covered
        bin A_data_max                                     35          1          -    Covered
        bin A_data_min                                     39          1          -    Covered
        default bin A_data_default                        155                     -    Occurred
    Coverpoint B                                       100.00%        100          -    Covered
        covered/total bins:                                 3          3          -
        missing/total bins:                                 0          3          -
        % Hit:                                         100.00%        100          -
        bin B_data_0                                       34          1          -    Covered
        bin B_data_max                                     24          1          -    Covered
        bin B_data_min                                     37          1          -    Covered
        default bin B_data_default                        163                     -    Occurred
    Coverpoint opcode                                  100.00%        100          -    Covered
        covered/total bins:                                 6          6          -
        missing/total bins:                                 0          6          -
        % Hit:                                         100.00%        100          -
        illegal_bin bins_invalid                            2                     -    Occurred
        bin bins_shift[SHIFT_OP]                           37          1          -    Covered
        bin bins_shift[ROT_OP]                             38          1          -    Covered
        bin bins_arith[ADD_OP]                             46          1          -    Covered
        bin bins_arith[MUL_OP]                             49          1          -    Covered
        bin bins_bitwise[OR_OP]                            41          1          -    Covered
        bin bins_bitwise[XOR_OP]                           45          1          -    Covered
```

9. Clear and neat QuestaSim waveform snippets showing the functionality of the design


Wave
File Edit View Add Format Tools Bookmarks Window Help
Wave - Default
/question_3_tb/clk
/question_3_tb/rst
/question_3_tb/A
/question_3_tb/B
/question_3_tb/cin
/question_3_tb/serial_in
/question_3_tb/red_op_A
/question_3_tb/red_op_B
/question_3_tb/bypass_A
/question_3_tb/bypass_B
/question_3_tb/direction
/question_3_tb/opcode
/question_3_tb/result
SHIFT_OP
XOR_OP
OR_OP
SHIFT_OP
ROT_OP
ADD_OP
Now
Cursor 1
9796 ns to 9868 ns
Activate Windows


Wave
File Edit View Add Format Tools Bookmarks Window Help
Wave - Default
/question_3_tb/clk
/question_3_tb/rst
/question_3_tb/A
/question_3_tb/B
/question_3_tb/cin
/question_3_tb/serial_in
/question_3_tb/red_op_A
/question_3_tb/red_op_B
/question_3_tb/bypass_A
/question_3_tb/bypass_B
/question_3_tb/direction
/question_3_tb/opcode
/question_3_tb/result
Now
Cursor 1
5363 ns to 5535 ns
Activate Windows

Q4

1. Testbench code

```
import mem_pkg::*;

module my_mem_tb;

  // ----------------------
  // DUT Signals
  // ----------------------
  logic clk;
  logic write;
  logic read;
  logic [7:0] data_in;
  logic [15:0] address;
  logic [7:0] data_out;

  // ----------------------
  // DUT Instantiation
  // ----------------------
  my_mem dut (
    .clk(clk),
    .write(write),
    .read(read),
    .data_in(data_in),
    .address(address),
    .data_out(data_out)
  );

  // ----------------------
  // Clock Generation
  // ----------------------
  initial begin
    clk = 0;
    forever #5 clk = ~clk;  // 10ns clock period
  end

```

```
35  // ----------------------
36  // Initial Block
37  // ----------------------
38  initial begin
39    // Initialize control signals
40    write = 0;
41    read = 0;
42    data_in = 0;
43    address = 0;
44
45    // 1. Generate stimulus
46    stimulus_gen();
47
48    // 2. Populate golden model
49    golden_model();
50
51    // 3. Write Phase
52    $display("Starting Write Phase...");
53    for (int i = 0; i < TESTS; i++) begin
54      @(negedge clk);
55      address = address_array[i];
56      data_in = data_to_write_array[i];
57      write = 1;
58      read = 0;
59
60      @(posedge clk); // Wait for write to complete
61    end
62    write = 0;
63
```

```
62      write = 0;
63
64      // 4. Read Phase and Self-Check
65      $display("Starting Read Phase...");
66      for (int i = 0; i < TESTS; i++) begin
67        @(negedge clk);
68        address = address_array[i];
69        write = 0;
70        read = 1;
71
72        @(posedge clk); // Wait for read to complete
73        check9Bits(address, {~^data_out, data_out}); // Includes parity check
74        data_read_queue.push_back({~^data_out, data_out});
75      end
76      read = 0;
77
78      // 5. Display Read Data from Queue
79      $display("Read Data Queue:");
80      while (data_read_queue.size() > 0) begin
81        static bit [8:0] temp = data_read_queue.pop_front();
82        $display("Data from queue: %0h", temp);
83      end
84
85      // 6. Final Report
86      $display("Simulation Completed.");
87      $display("Correct Count = %0d", correct_count);
88      $display("Error Count   = %0d", error_count);
89
90      $stop;
91    end
92
93  endmodule
```

2. Package code

```
package mem_pkg;

  // Number of test operations
  localparam int TESTS = 100;

  // ----------------------
  // Data Structures
  // ----------------------
  // Queues arrays for stimulus
  bit [15:0] address_array[$];          // Random addresses
  bit [7:0]  data_to_write_array[$];    // Random data to be written

  // Associative array for expected read values (includes parity)
  bit [8:0] data_read_expect_assoc[int];

  // Queue for actual data read from DUT
  bit [8:0] data_read_queue[$];

  // Error and correct counters
  int error_count = 0;
  int correct_count = 0;

  // ----------------------
  // Task: Stimulus Generation
  // ----------------------
  task automatic stimulus_gen();
    $display("Generating Stimulus...");
    address_array.delete();
    data_to_write_array.delete();

    for (int i = 0; i < TESTS; i++) begin
      address_array.push_back($urandom_range(0, 65535));
      data_to_write_array.push_back($urandom_range(0, 255));
    end
  endtask
```

```
  // ----------------------
  // Task: Golden Model
  // ----------------------
  task automatic golden_model();
    $display("Populating Golden Model...");
    data_read_expect_assoc.delete();

    for (int i = 0; i < TESTS; i++) begin
      bit parity = ~^data_to_write_array[i]; // Even parity
      data_read_expect_assoc[address_array[i]] = {parity, data_to_write_array[i]};
    end
  endtask

  // ----------------------
  // Task: Check 9-bit data
  // ----------------------
  task automatic check9Bits(input bit [15:0] addr, input bit [8:0] actual_data);
    if (actual_data !== data_read_expect_assoc[addr]) begin
      $display("ERROR: Addr=%0h Expected=%0h Got=%0h", addr, data_read_expect_assoc[addr], actual_data);
      error_count++;
    end
    else begin
      $display("CORRECT: Addr=%0h Data=%0h", addr, actual_data);
      correct_count++;
    end
  endtask

endpackage
```

3. Design code

```
module my_mem (
  input  logic        clk,
  input  logic        write,
  input  logic        read,
  input  logic [7:0]  data_in,
  input  logic [15:0] address,
  output logic [7:0]  data_out
);

  // Memory Declaration: 9-bit wide, 64K depth
  logic [8:0] mem_array [0:65535];

  always_ff @(posedge clk) begin
    if (write) begin
      // Even parity calculation
      mem_array[address] <= {~^data_in, data_in};
    end
    else if (read) begin
      data_out <= mem_array[address][7:0]; // Output only the 8-bit data
    end
  end

endmodule

```

4. Report any bugs detected in the design and fix them

5. Snippet to your verification plan document

Label	Design Requirement Description	Stimulus Generation	Functional Coverage	Functionality Check
MEM_1	RAM should **write data correctly** on positive edge of clk when write = 1.	Random addresses and random data values using address_array and data_to_write_array.	-	Checker ensures data stored in RAM matches the written value plus parity bit using check9Bits task.
MEM_2	RAM should **read data correctly** on positive edge of clk when read = 1.	Random addresses generated from address_array.	-	Checker compares data_out with expected data stored in data_read_expect_assoc.
MEM_3	**Write operation has higher priority** than read when both read and write are high at the same clock edge.	Directed test where both read and write are asserted together.	-	Checker ensures write happens and read is ignored when both are high.
MEM_4	**Read and write cannot happen at the same time** — only one operation allowed per clock cycle.	Directed test toggling read and write.	-	Assertion or checker verifies that both are not active simultaneously.
MEM_5	**Even parity bit** must be calculated correctly and stored as the MSB (9th bit) when data is written.	Random data writes, verify parity calculation ~^data_in.	-	check9Bits task ensures parity matches even parity rule: parity_bit = ~^data_in.
MEM_6	Data should persist at the given address after multiple operations (no data corruption).	Random writes followed by random reads to same addresses.	-	Checker ensures previously written data is retained unless overwritten.
MEM_7	Verify **memory depth** supports 64K addresses (16-bit address space).	Random addresses covering full range from 0 to 65535.	-	Checker ensures all addresses can be accessed and data matches expectations.
MEM_8	**Output data queue** should store and display all read values in order.	Capture read data into data_read_queue.	-	At end of simulation, pop and display values using pop_front to confirm order of reads is correct.
MEM_9	Verify **error counting** works correctly.	Introduce intentional mismatches (e.g., flip a bit in expected data).	-	Verify error_count increments on mismatch and correct_count increments on successful read.
MEM_10	Reset behavior: confirm no writes or reads occur when simulation starts with all signals stable.	Directed initial condition where read = 0 and write = 0.	-	Checker ensures no change in memory or outputs when idle.

6. Do file

```
vlib work
vlog my_mem.sv  mem_pkg.sv my_mem_tb.sv  +cover -covercells
vsim -voptargs=+acc work.my_mem_tb -cover
add wave *
coverage save mem.ucdb -onexit
run -all
```

7. Clear and neat QuestaSim waveform snippets showing the functionality of the design